# When a polygon changes

Geospatial Data Processing at Scale

- A Zadu Case Study –

Jeroen Stiers

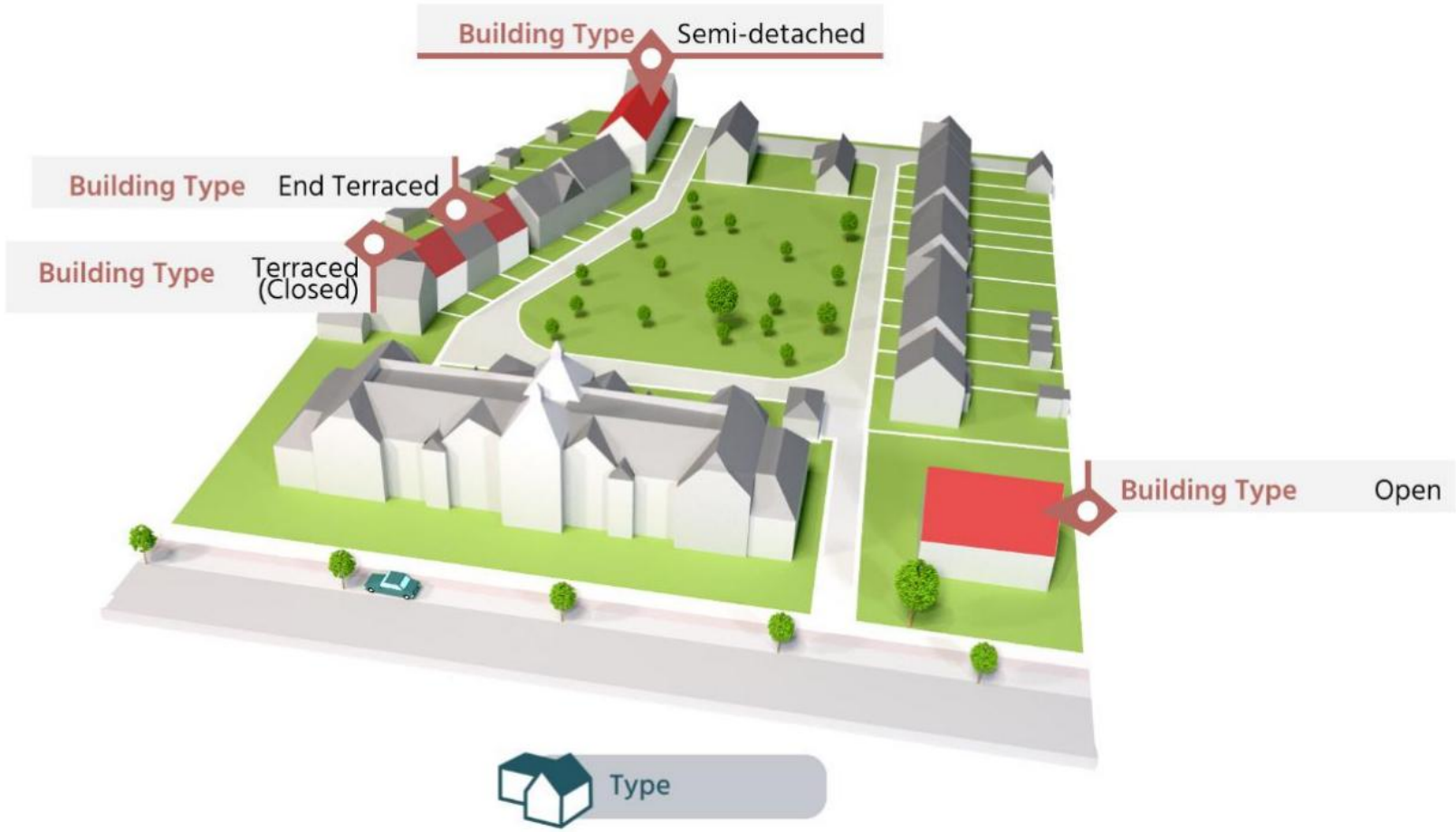

970 m³
Gable roof
Built in 1960
Low flood risk zone
Solar Panels
Library
Value: 435.000 €
25 % paved
High Street, 5

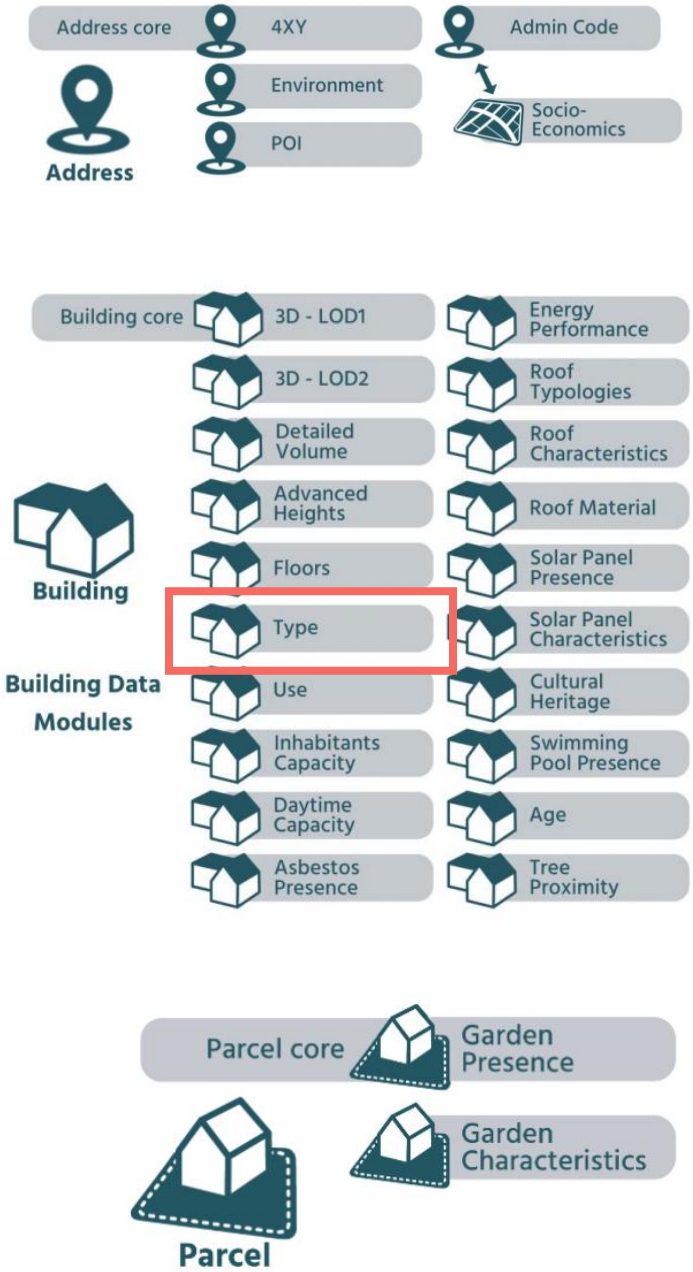


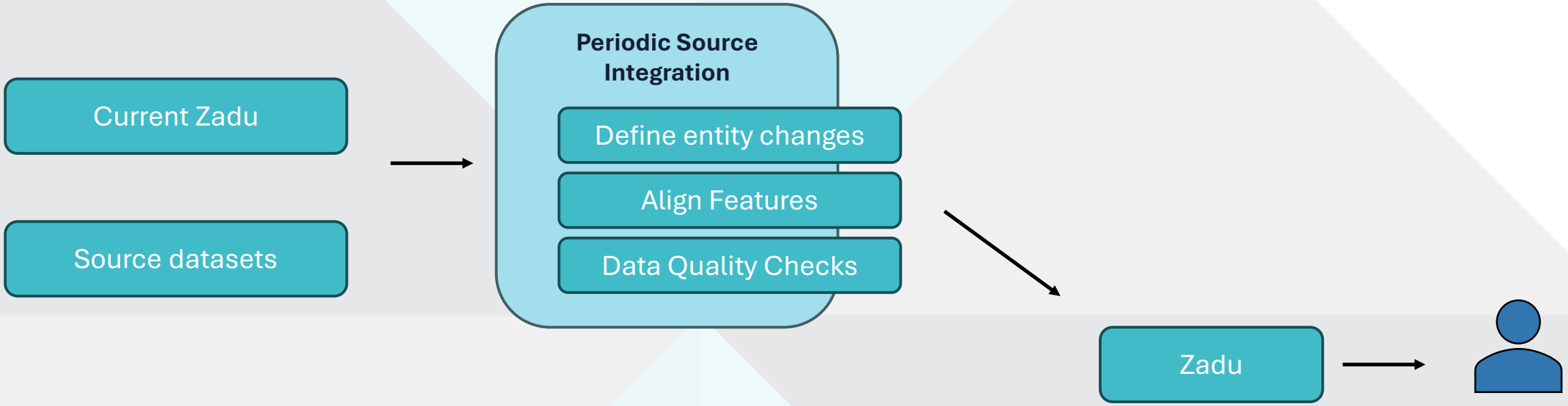

Address core
4XY
Admin Code
Environment
Socio-Economics
POI
Address

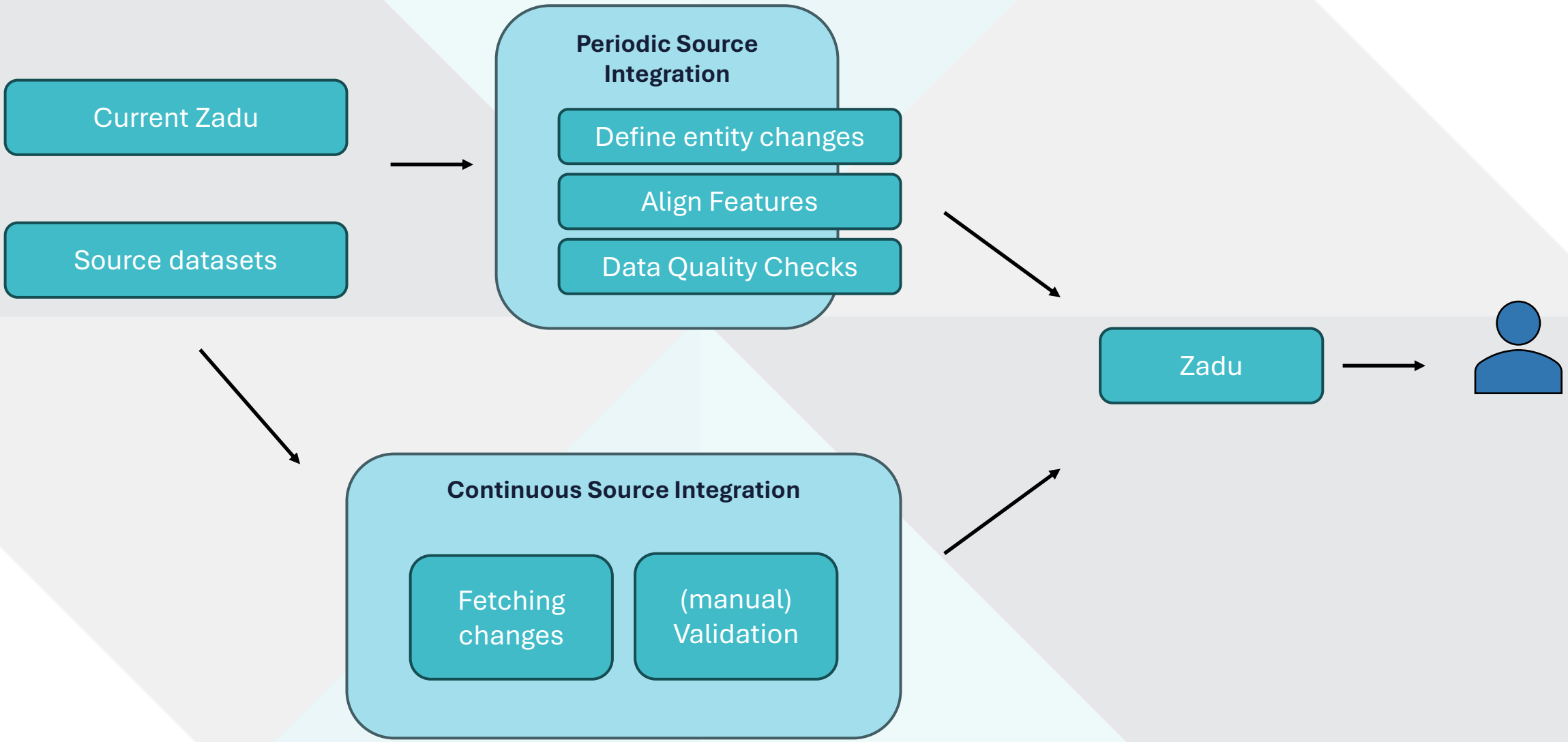
Building core
3D - LOD1
Energy Performance
3D - LOD2
Roof Typologies
Detailed Volume
Roof Characteristics
Advanced Heights
Roof Material
Floors
Solar Panel Presence
Type
Solar Panel Characteristics
Use
Cultural Heritage
Inhabitants Capacity
Swimming Pool Presence
Daytime Capacity
Age
Asbestos Presence
Tree Proximity
Building
Building Data Modules

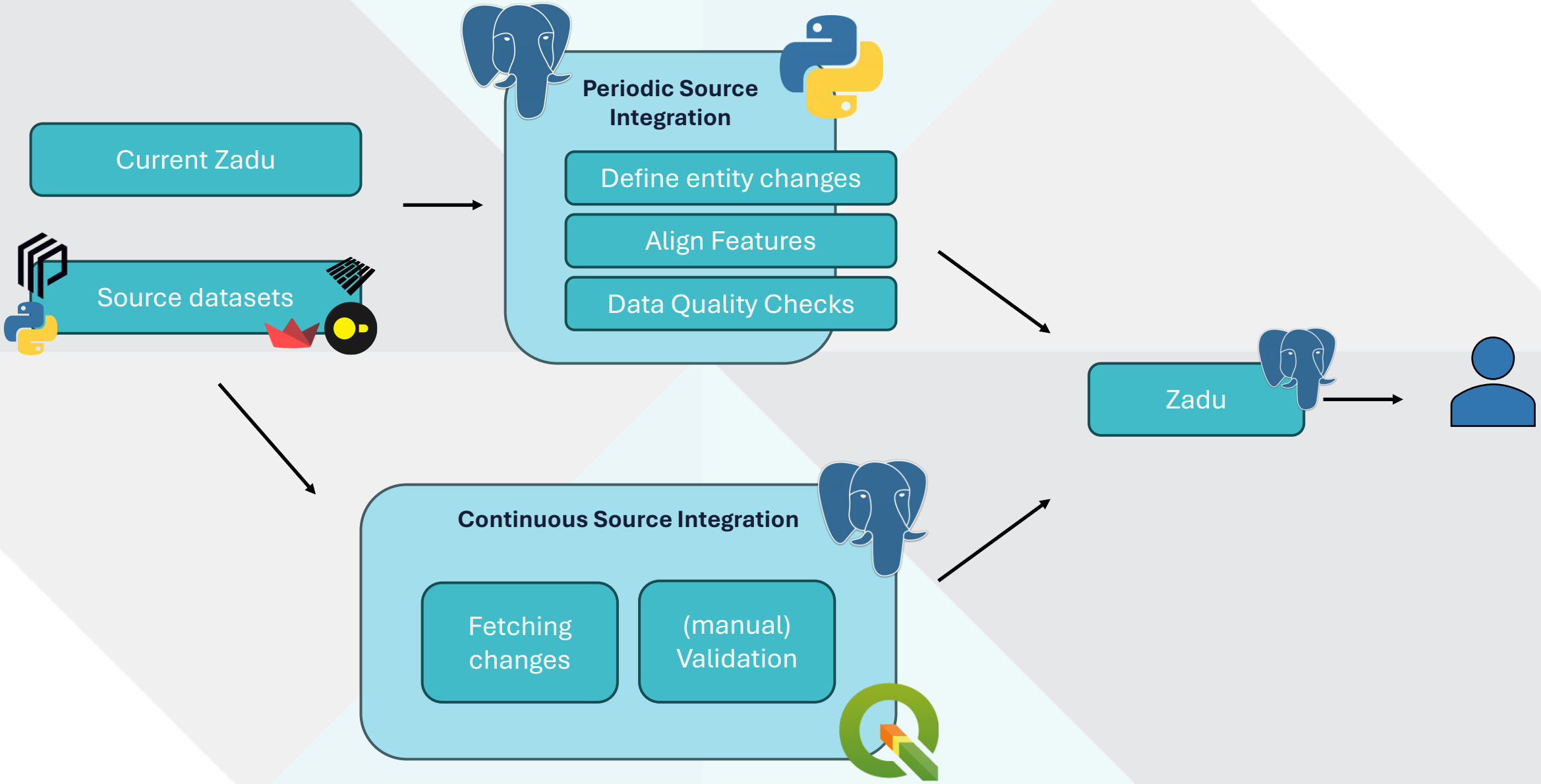
Parcel core
Garden Presence
Garden Characteristics
Parcel

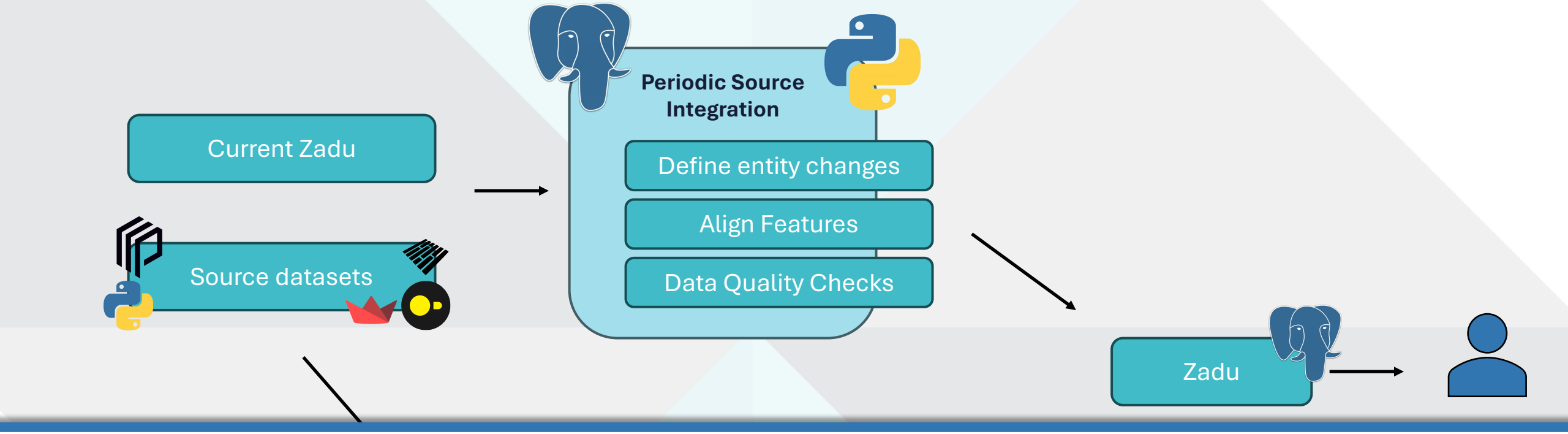

Building Type Semi-detached
Building Type End Terraced
Building Type Terraced (Closed)
Building Type Open
Type


Address core
4XY
Admin Code
Environment
Socio-Economics
POI
Address
Building core
3D - LOD1
Energy Performance
3D - LOD2
Roof Typologies
Detailed Volume
Roof Characteristics
Advanced Heights
Roof Material
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Solar Panel Presence
Type
Solar Panel Characteristics
Building
Building Data Modules
Use
Cultural Heritage
Inhabitants Capacity
Swimming Pool Presence
Daytime Capacity
Age
Asbestos Presence
Tree Proximity
Parcel core
Garden Presence
Garden Characteristics
Parcel


Current Zadu
Source datasets
Periodic Source Integration
Define entity changes
Align Features
Data Quality Checks
Zadu

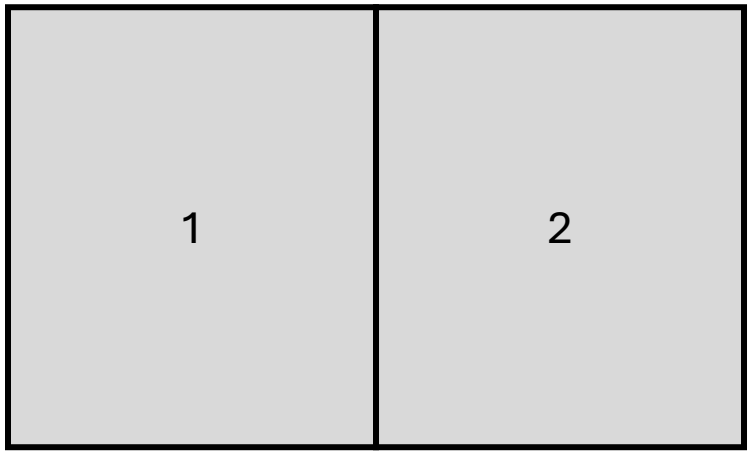

Current Zadu
Source datasets
Periodic Source Integration
Define entity changes
Align Features
Data Quality Checks
Continuous Source Integration
Fetching changes
(manual) Validation
Zadu

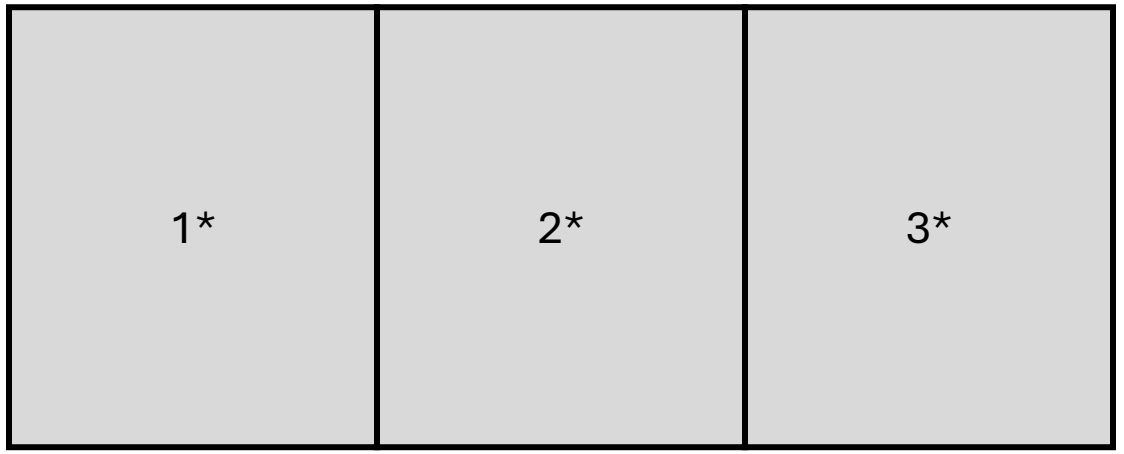

Current Zadu
Source datasets
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Define entity changes
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(manual) Validation
Zadu




| | Addresses | Buildings | Cadastral Parcels |
|---|---|---|---|
| **Belgium** | ~ 7M | ~ 8M | ~ 9M |
| **Luxembourg** | ~ 180K | ~ 240K | ~ 715K |
| **The Netherlands** | ~ 10M | ~ 11M | ~ 9M |

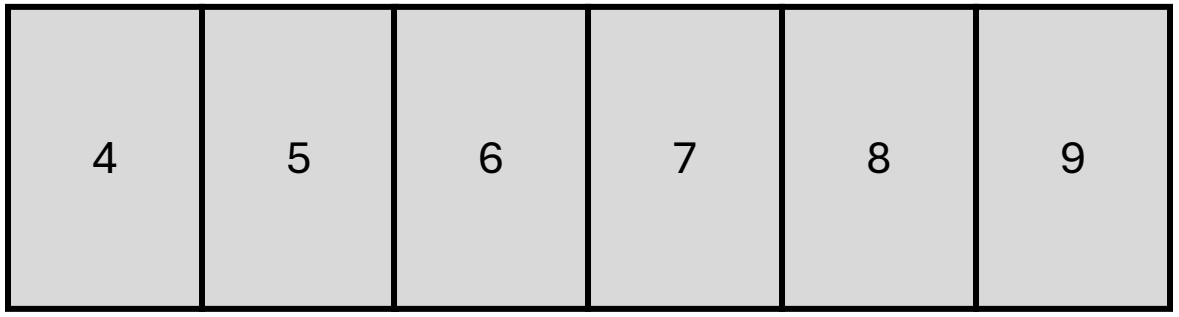
**Should run anywhere, for every country**

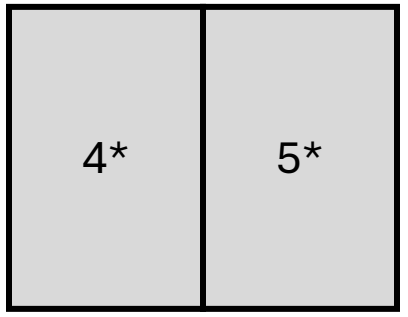


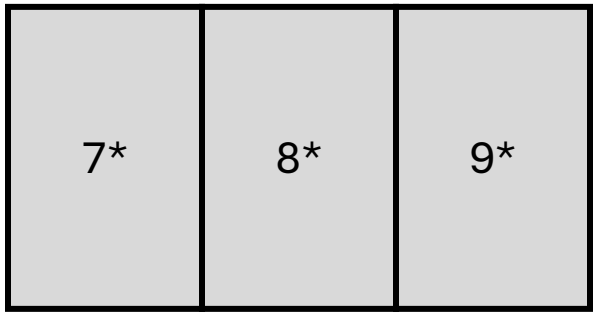



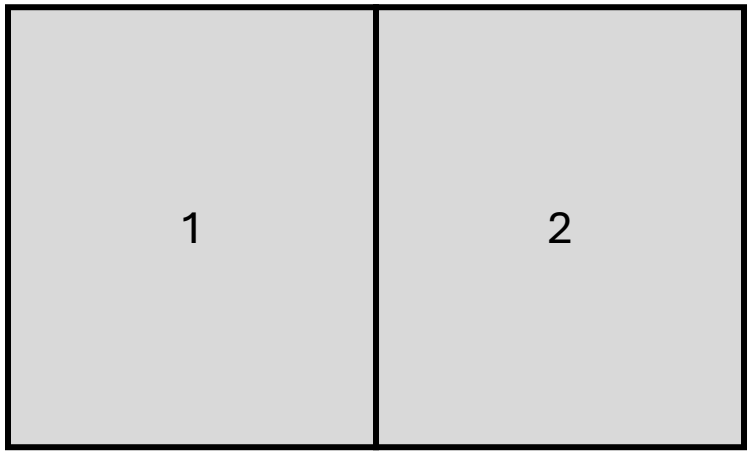

**Zadu**

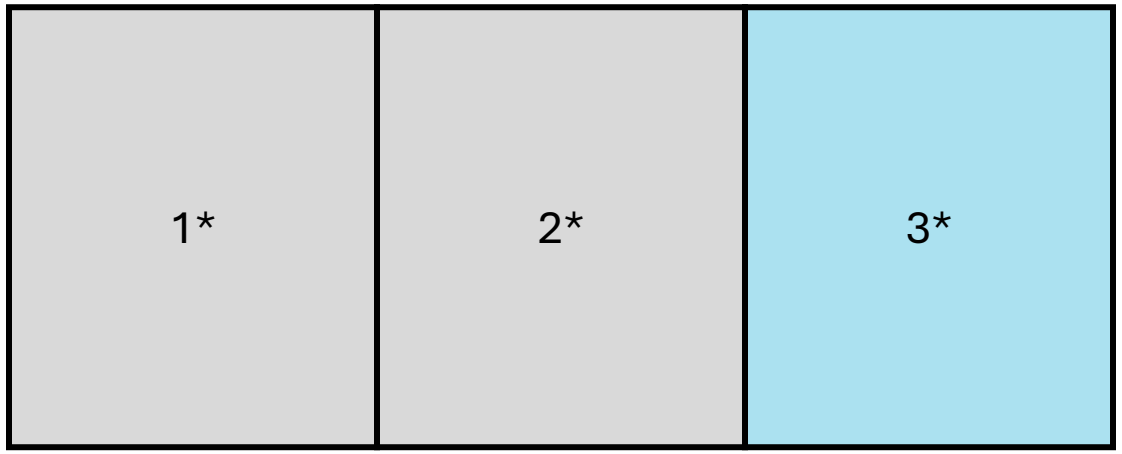
**Latest version source data**



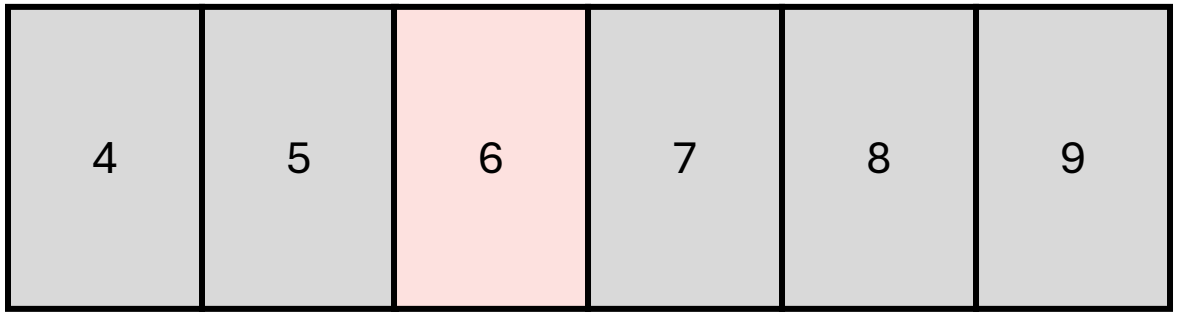



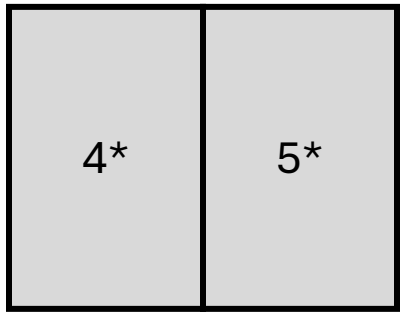


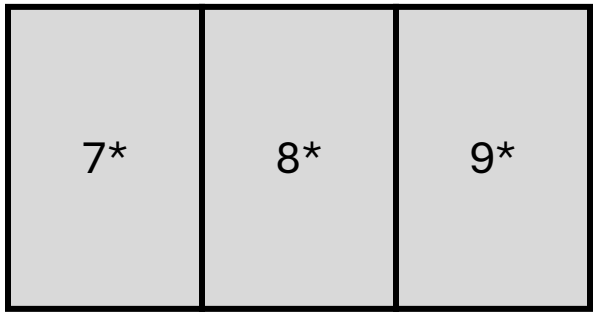




**Zadu**

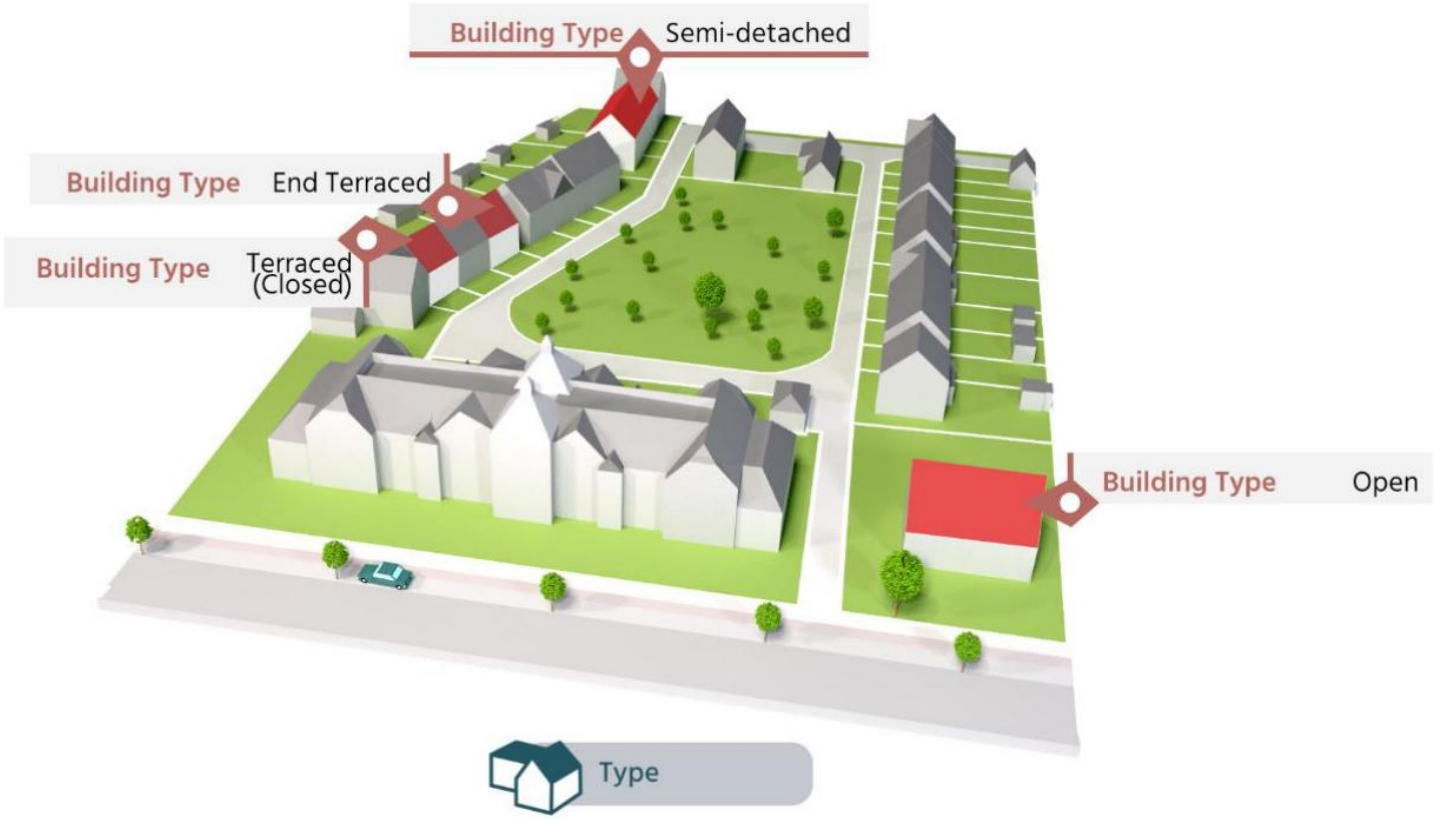
**Latest version source data**



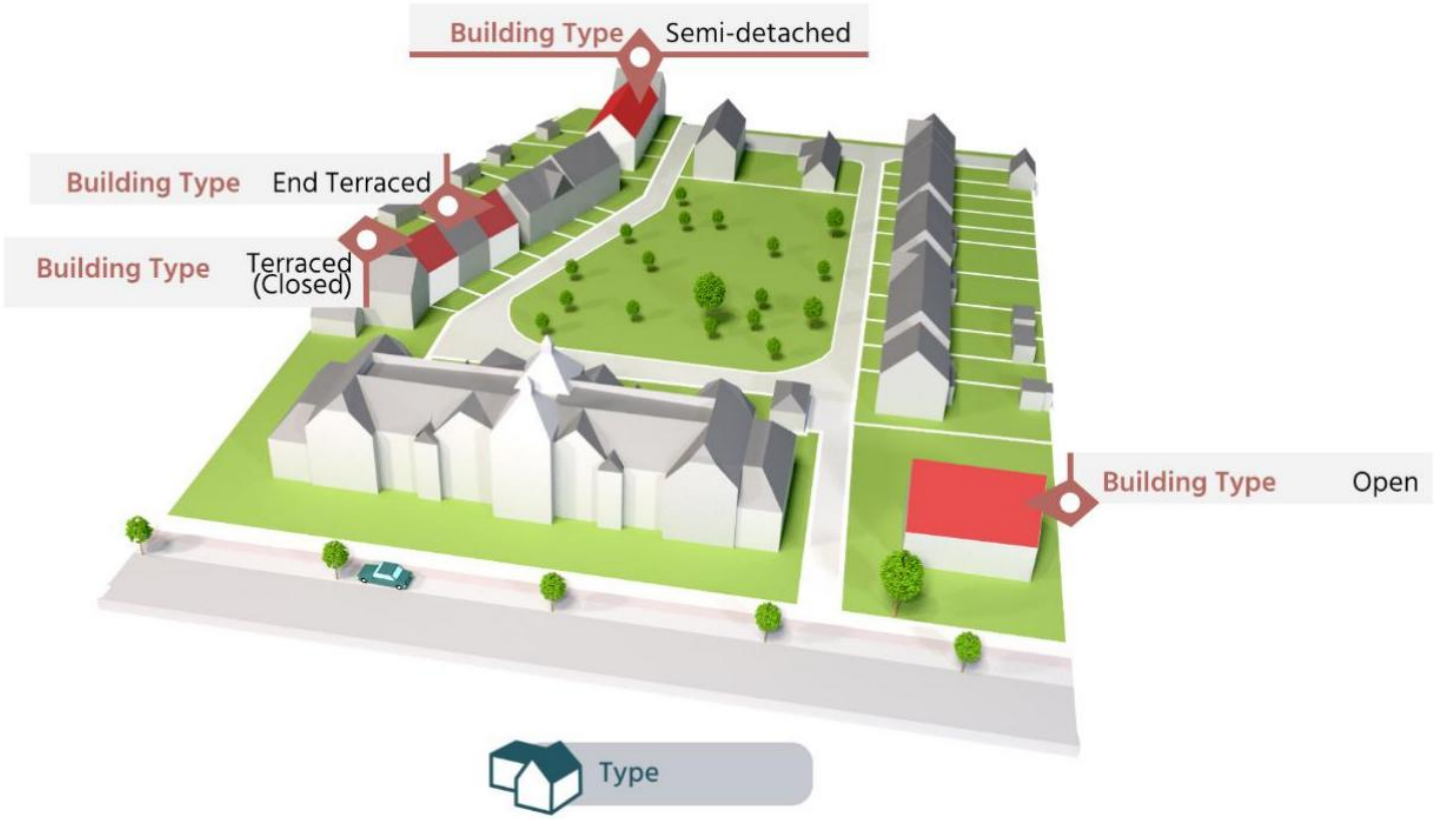



Building




| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |



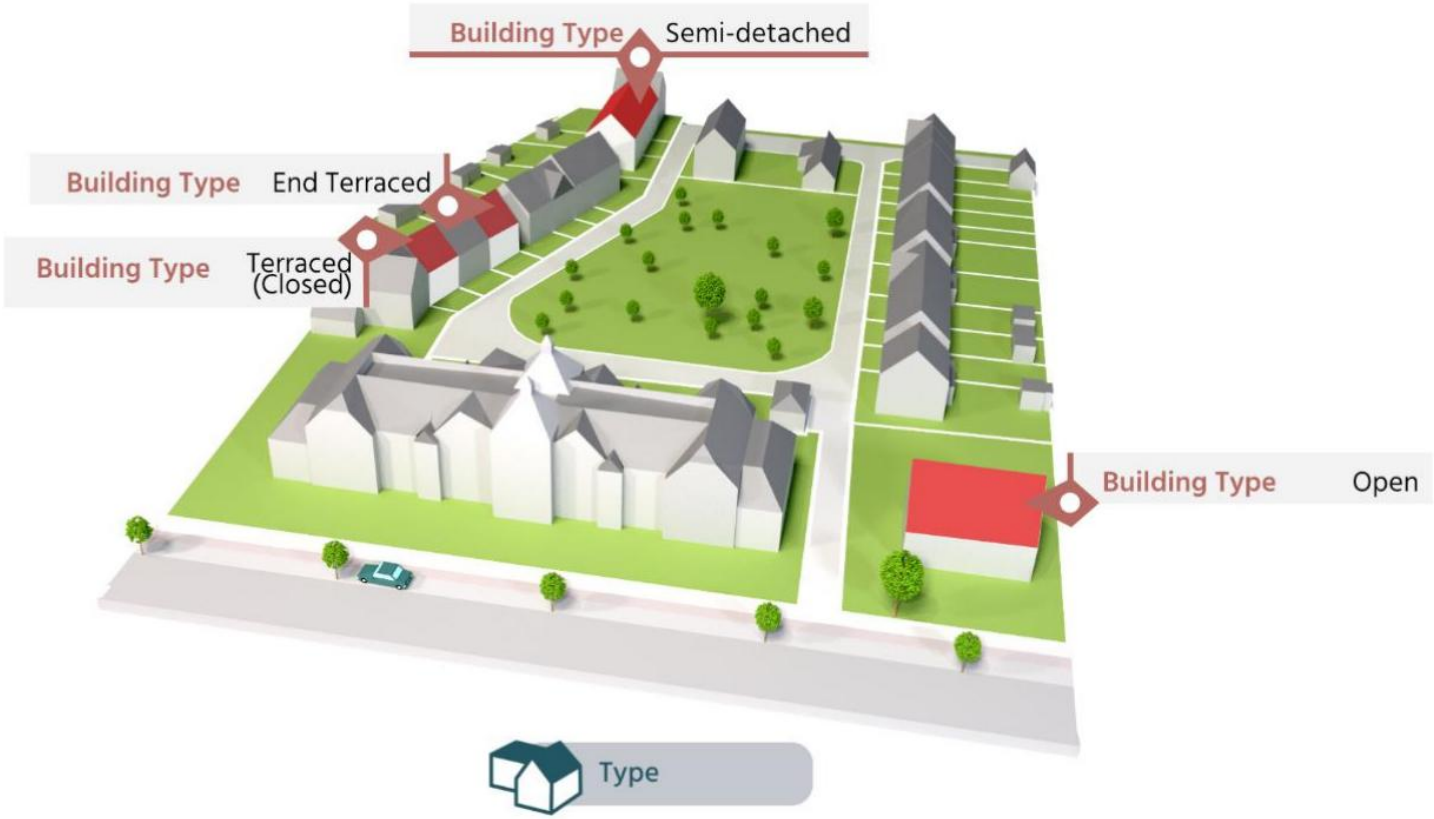

Building

Building Block

| 1 | 2 | 3 | 4 |
|---|---|---|---|

| 5 | 6 | 7 | 8 |
|---|---|---|---|

| 9 |
|---|




Building

Building Block

| 1 | 2 | 3 | 4 |
|---|---|---|---|

| 5 | 6 | 7 | 8 |
|---|---|---|---|

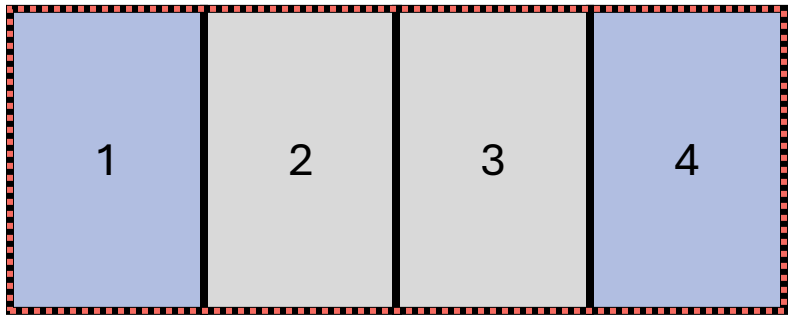
| 9 |
|---|

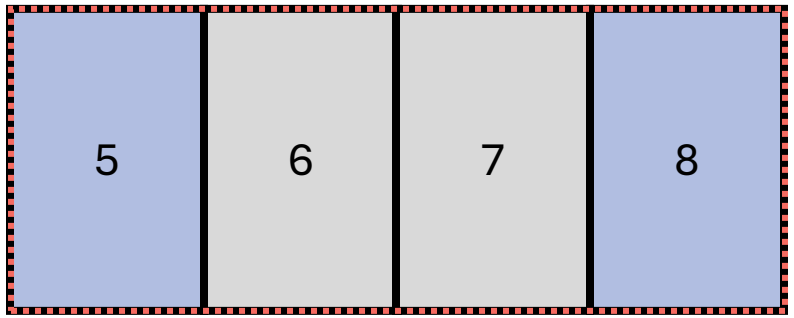
# Should run anywhere, for every country

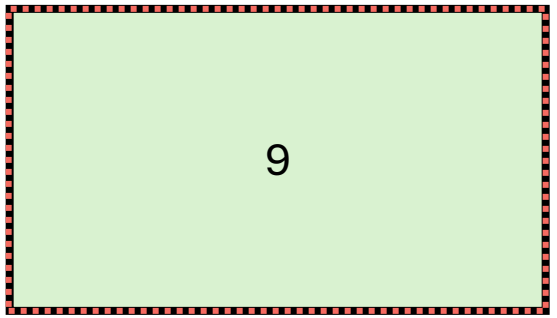



Building

Building Block



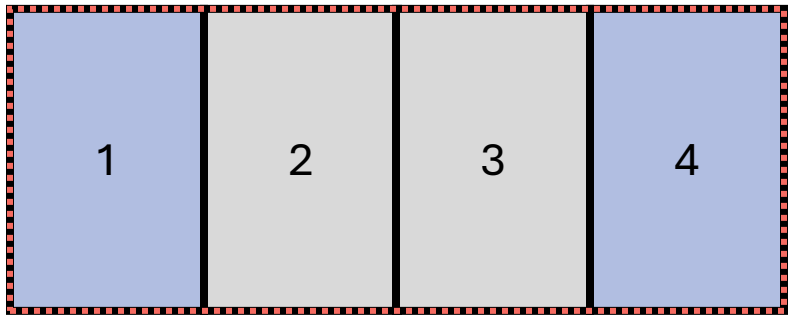
# Should run anywhere, for every country

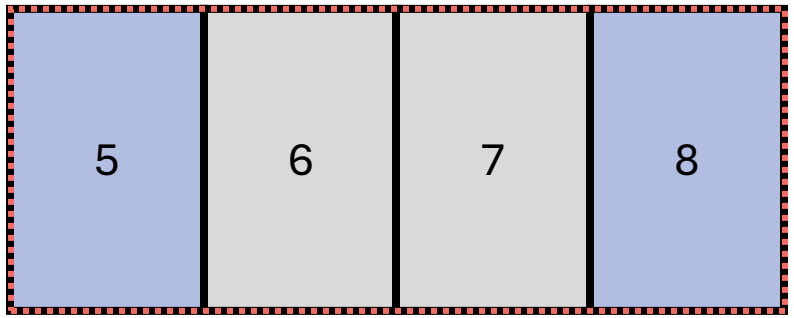
## Using a ‘tiled’ approach

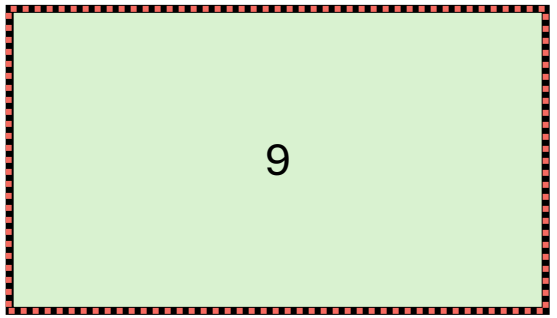



Building

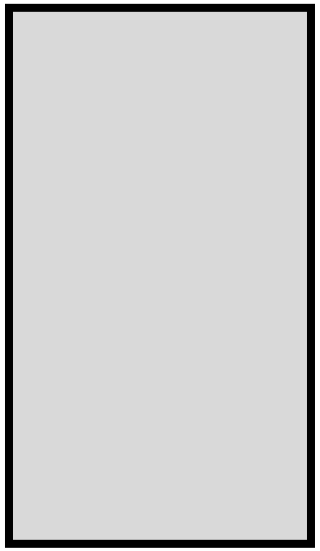
Building Block

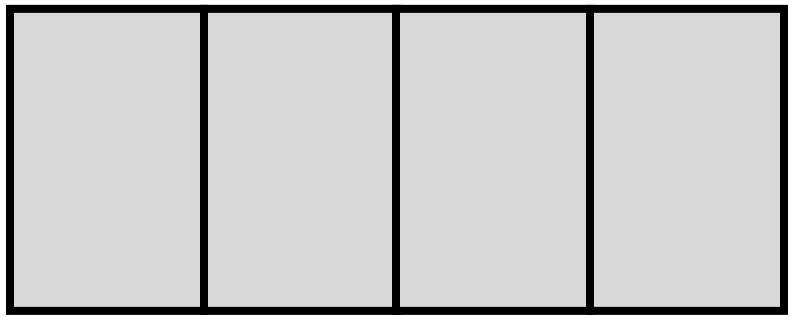


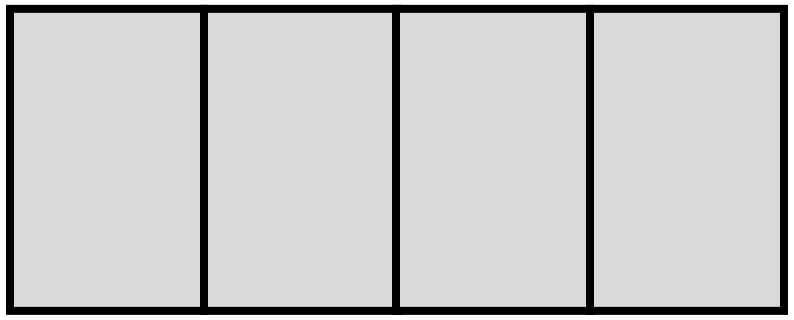



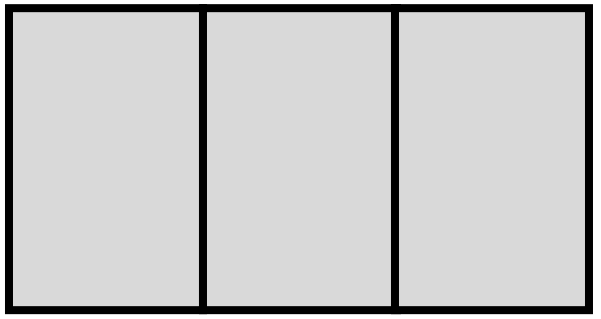



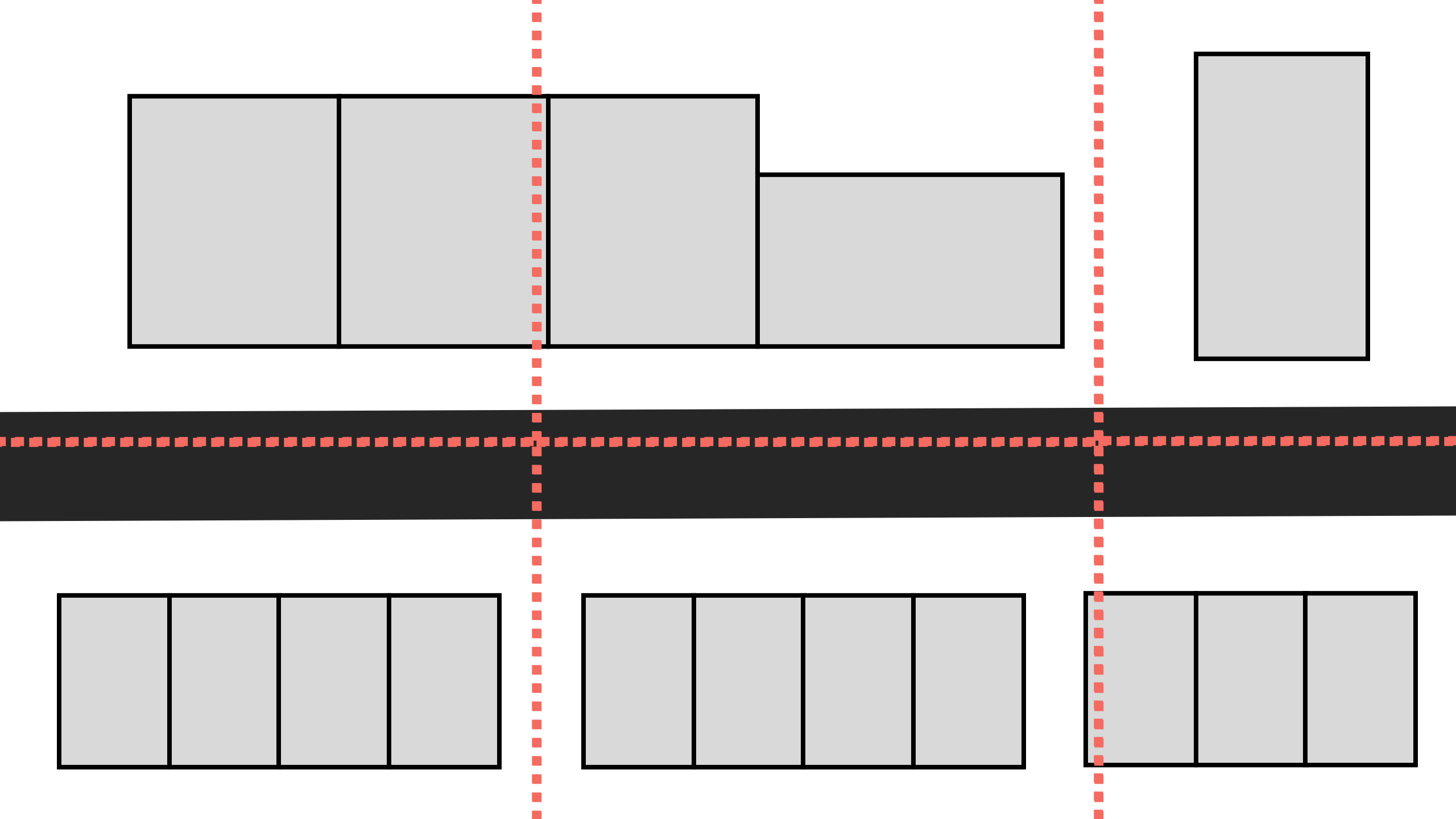


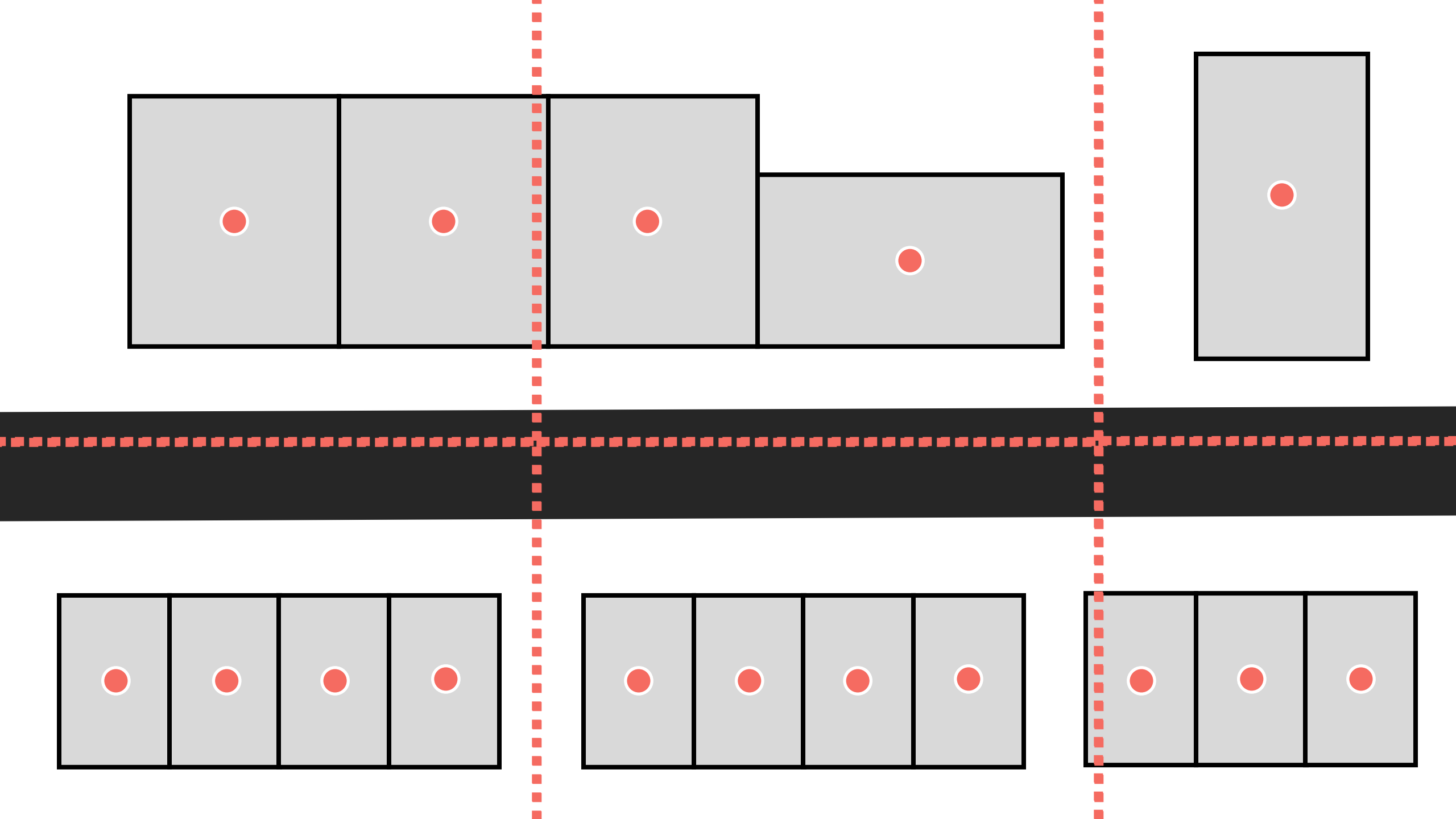


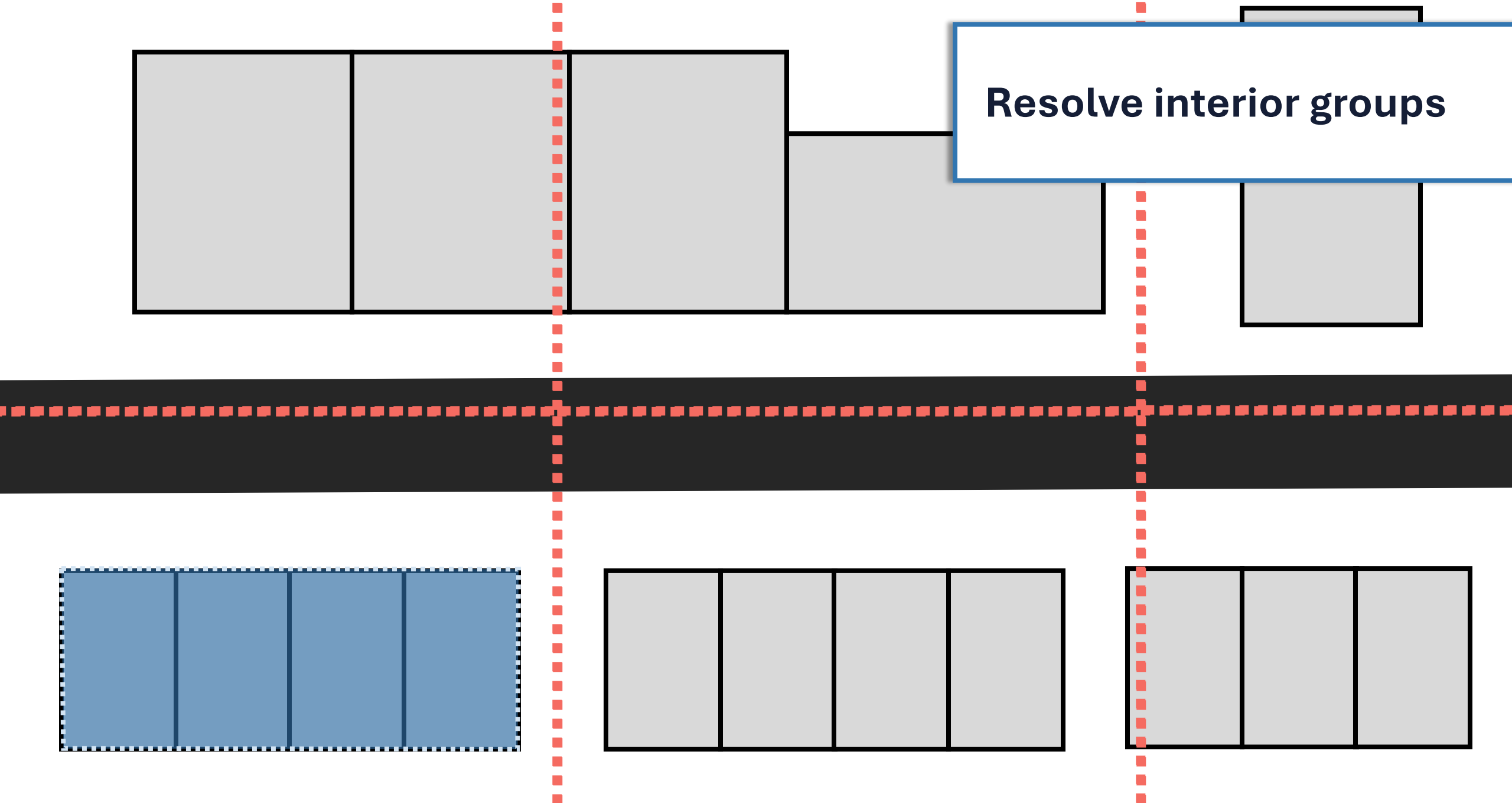


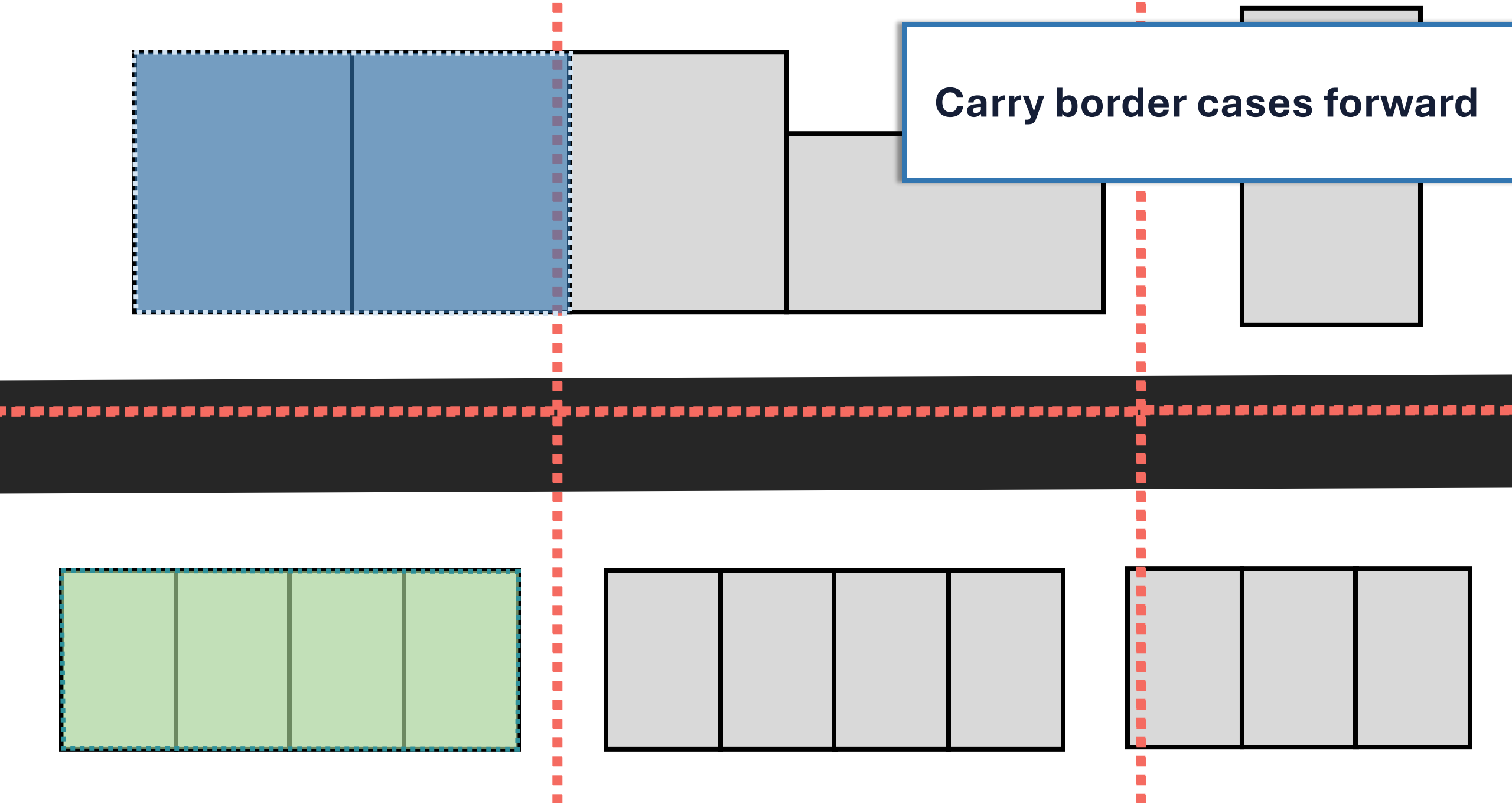


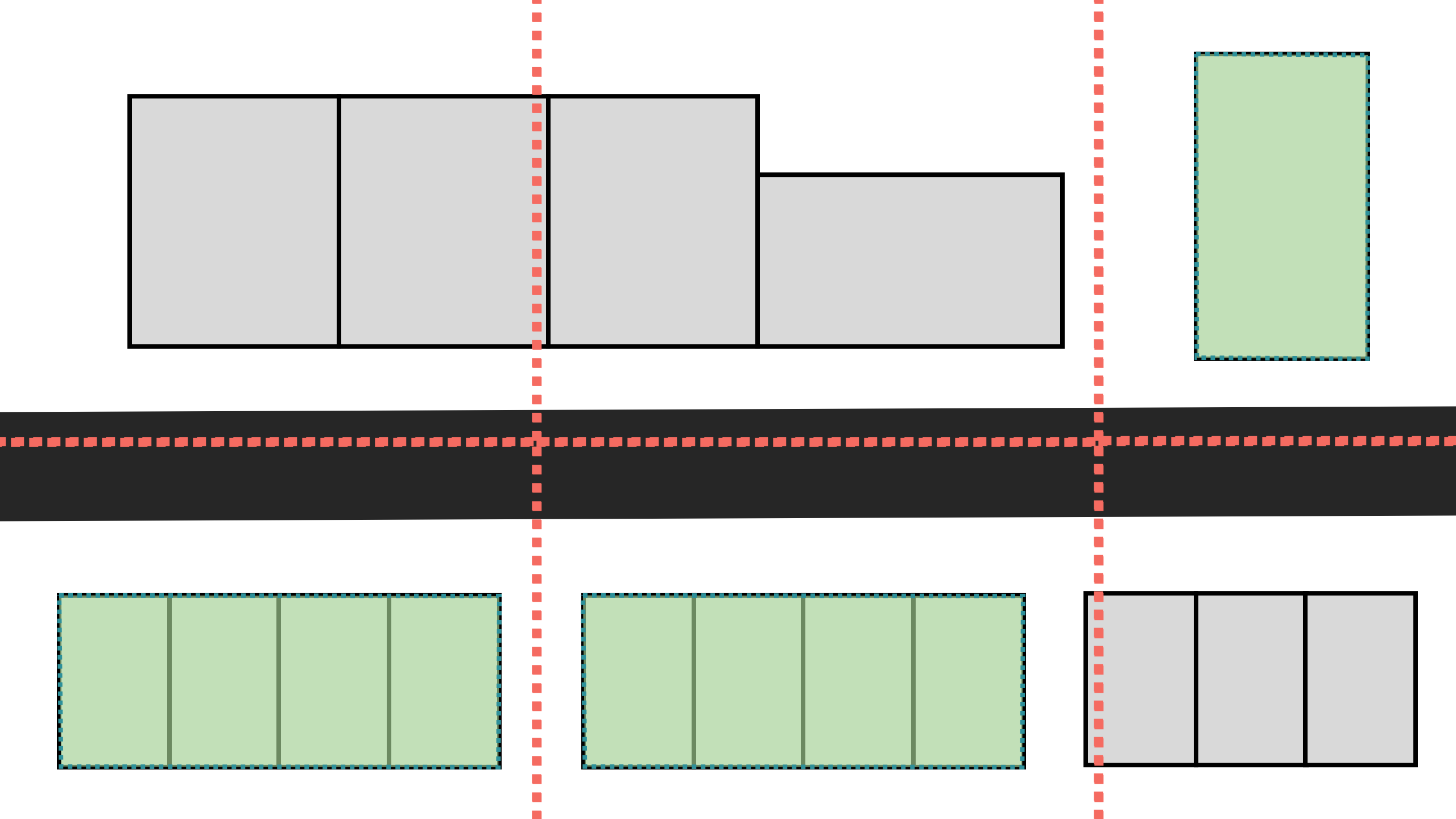


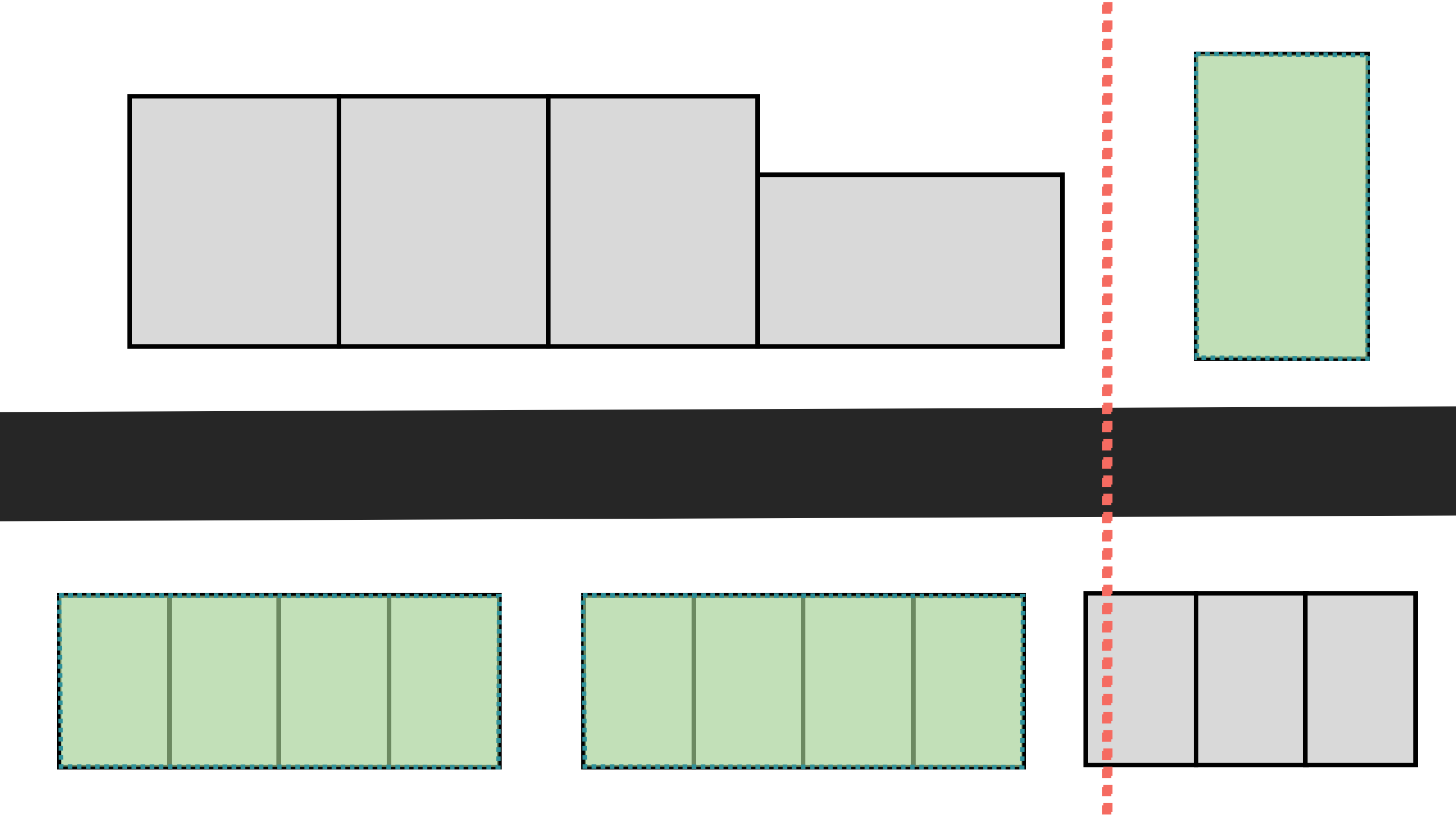


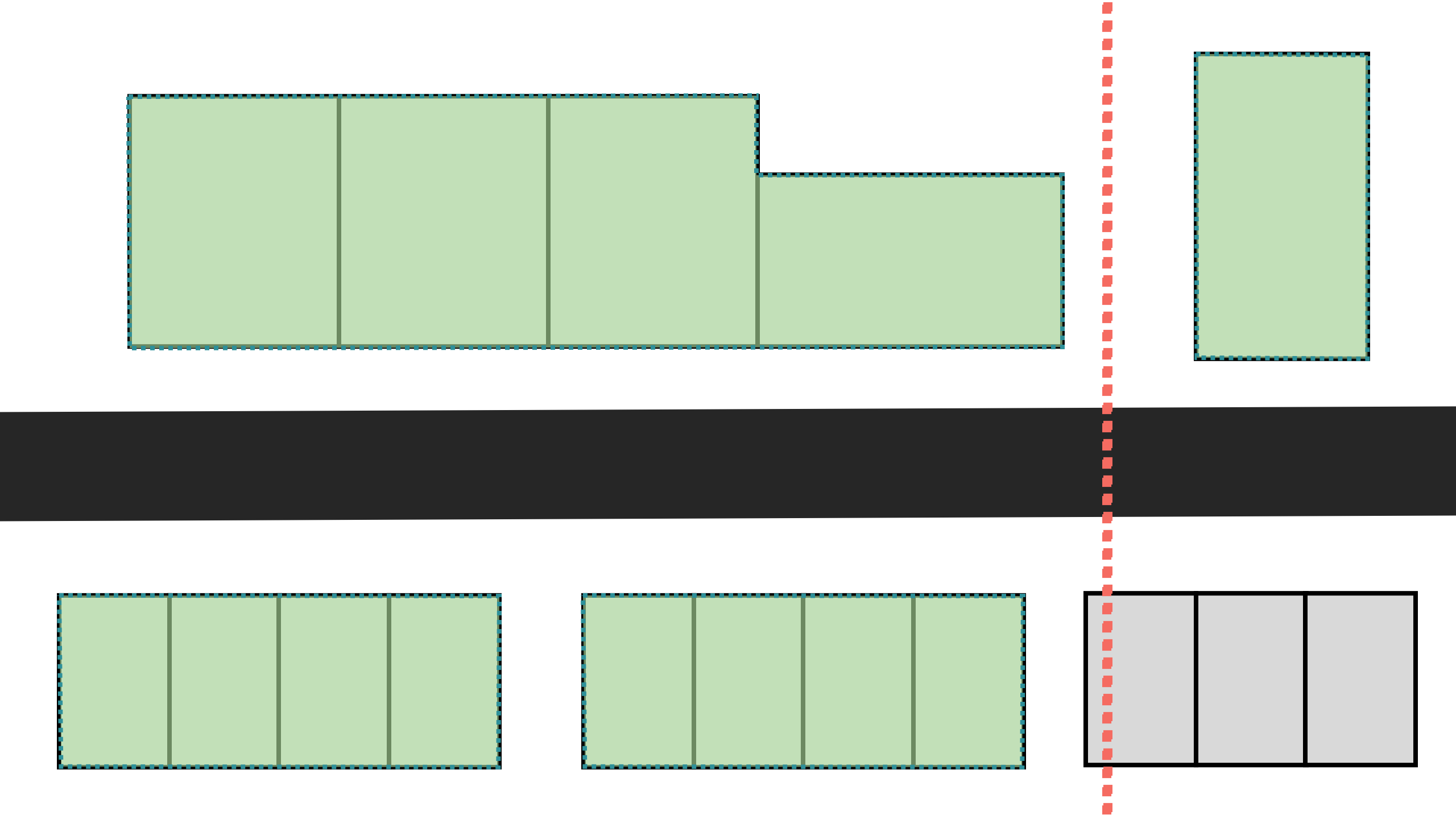


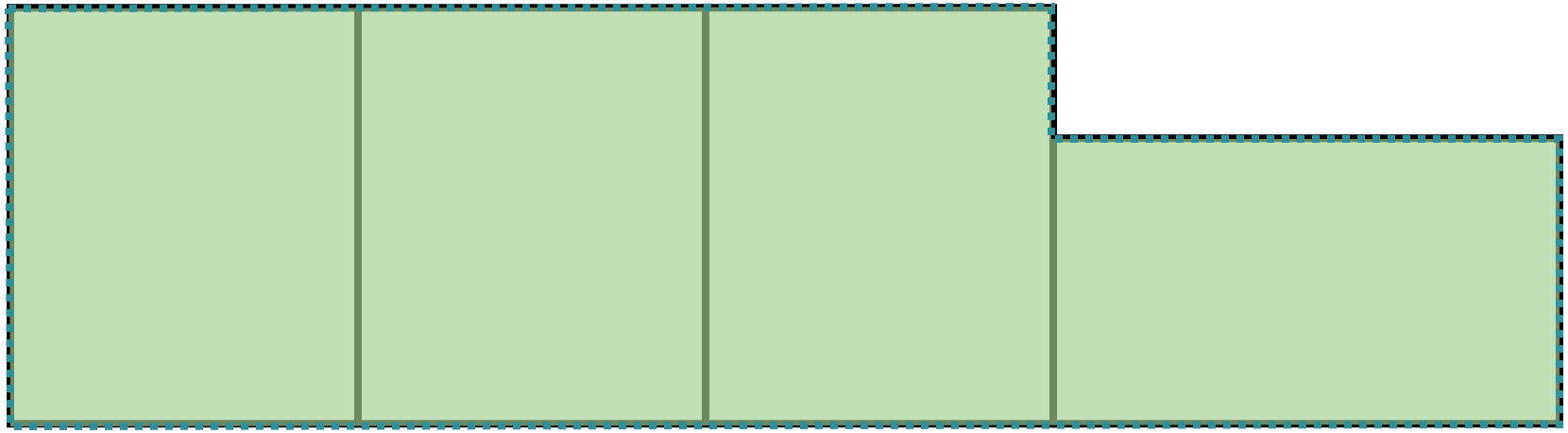

Resolve interior groups

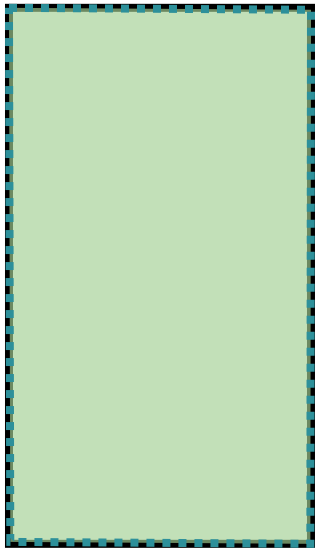

Carry border cases forward

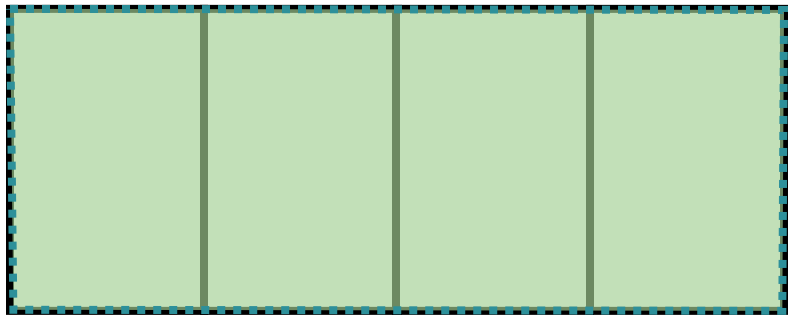


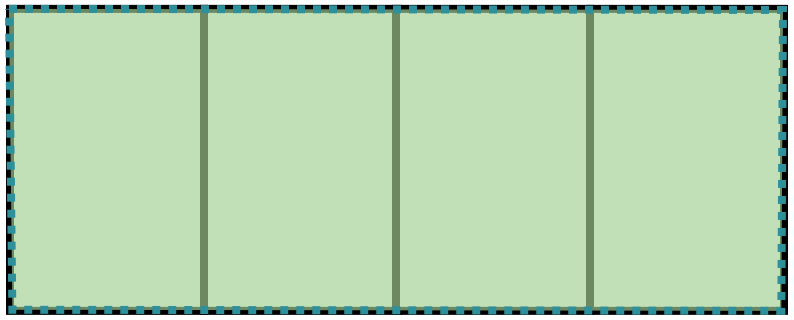


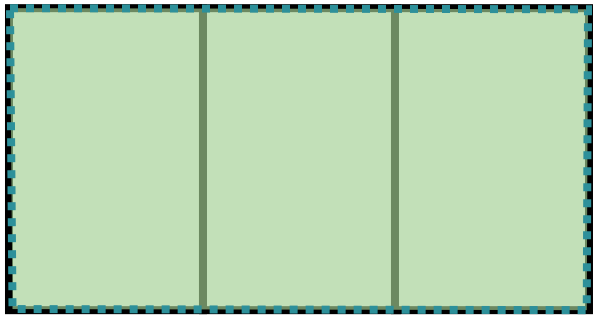


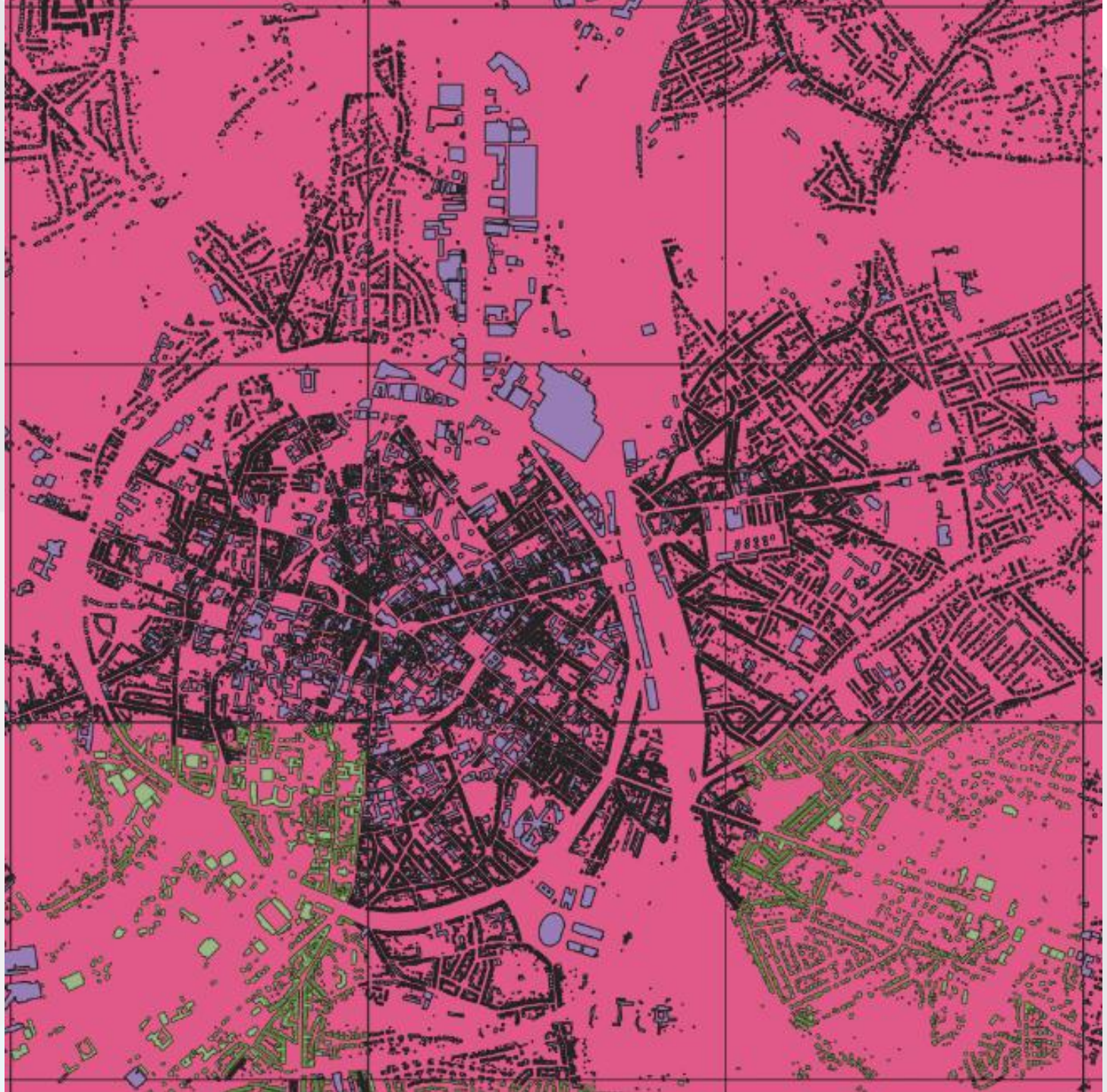




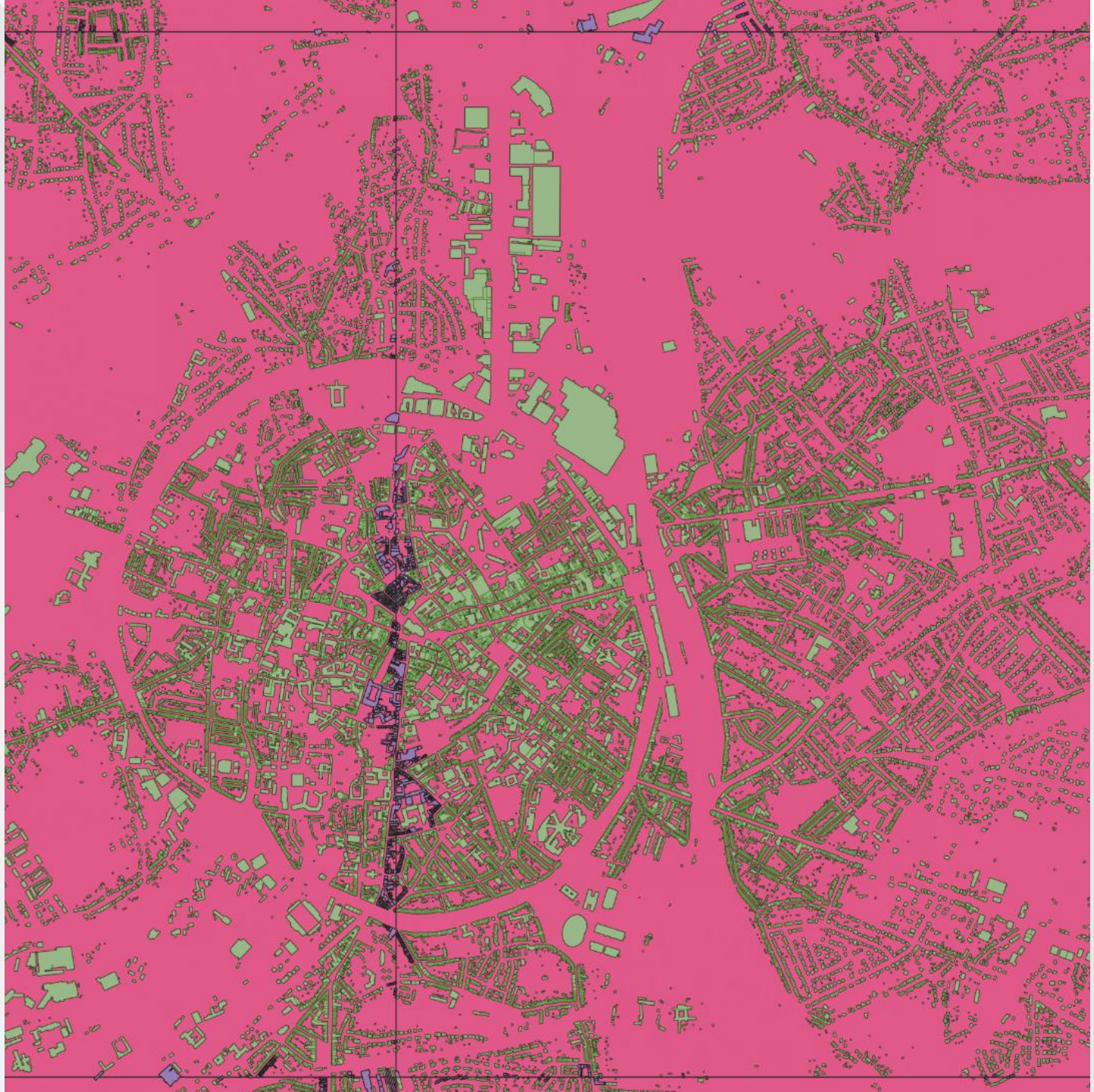














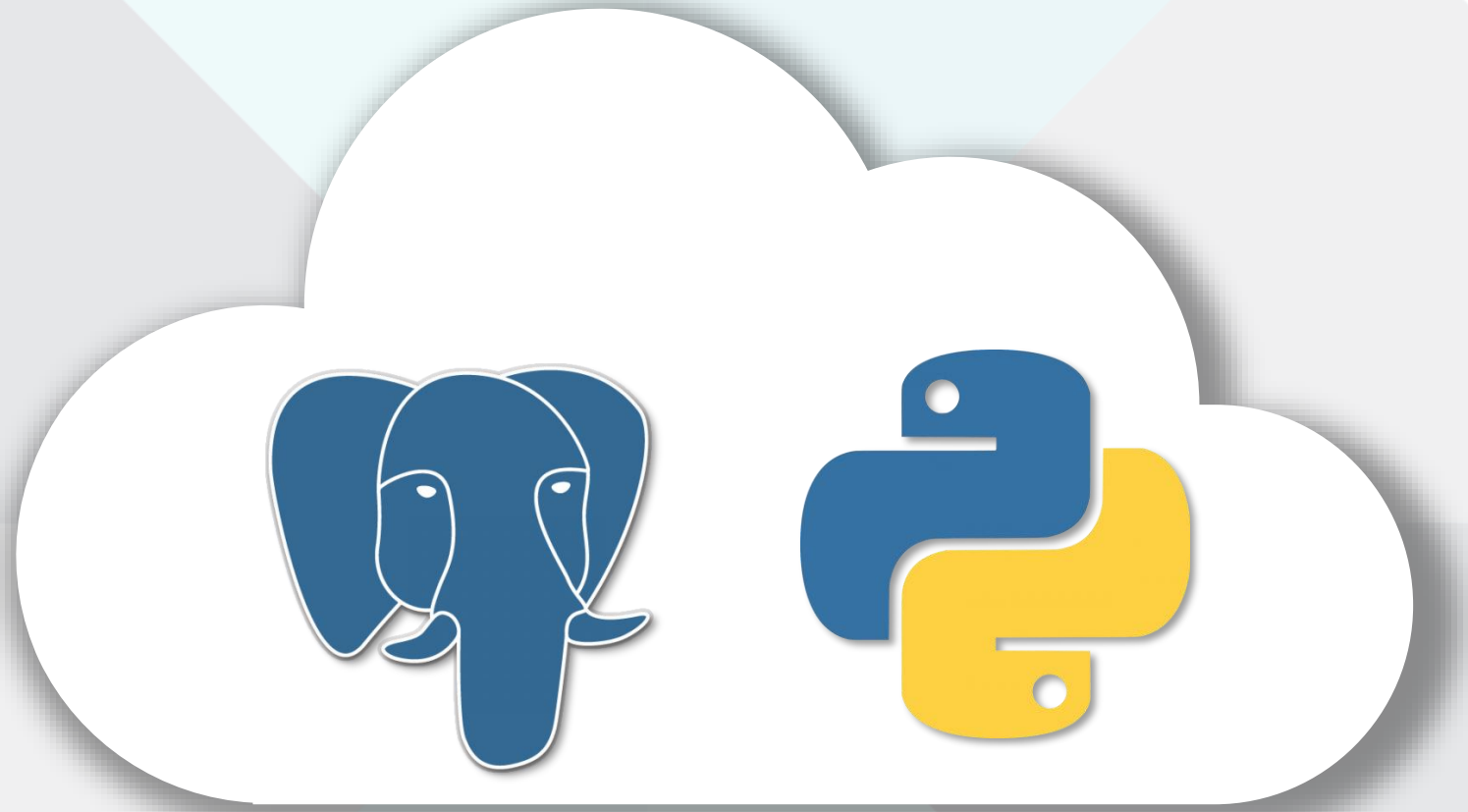
```python
database = PostgresqlConnection("...")

spatial_tiler = PolygonTiler(
    database=database,
    tablename="buildings",
    id_field="id"
    tile_size=5000
).run()

spatial_tiles = spatial_tiler.get_tiles()

database.run_query_tiled(
    query="""
        INSERT INTO buildingblocks (block_id, building_id)
        SELECT ...
        FROM buildings b
        INNER JOIN buildings_tiled bt ON bt.id = b.id
        WHERE bt.tile_id = {tile_id}
    """,
    tiles=spatial_tiles,
    tile_id_text="{tile_id}"
)
```

zadu

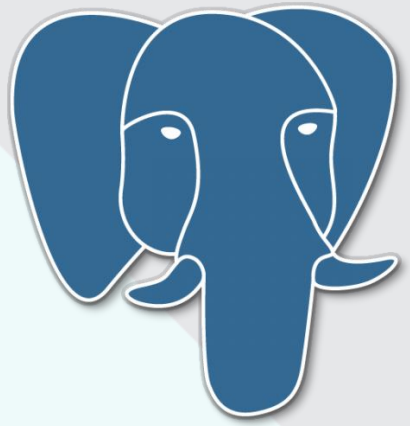


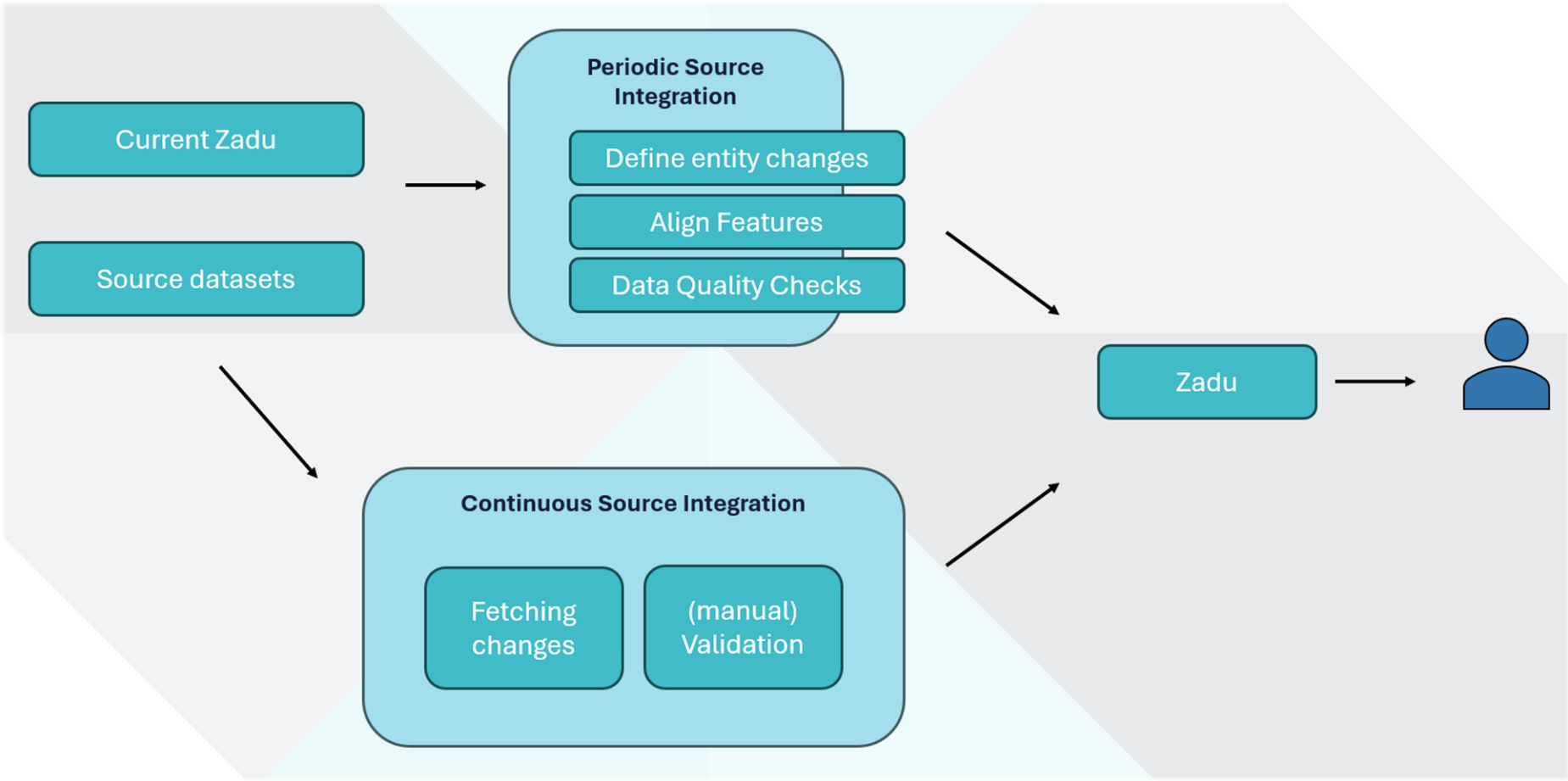
zadu

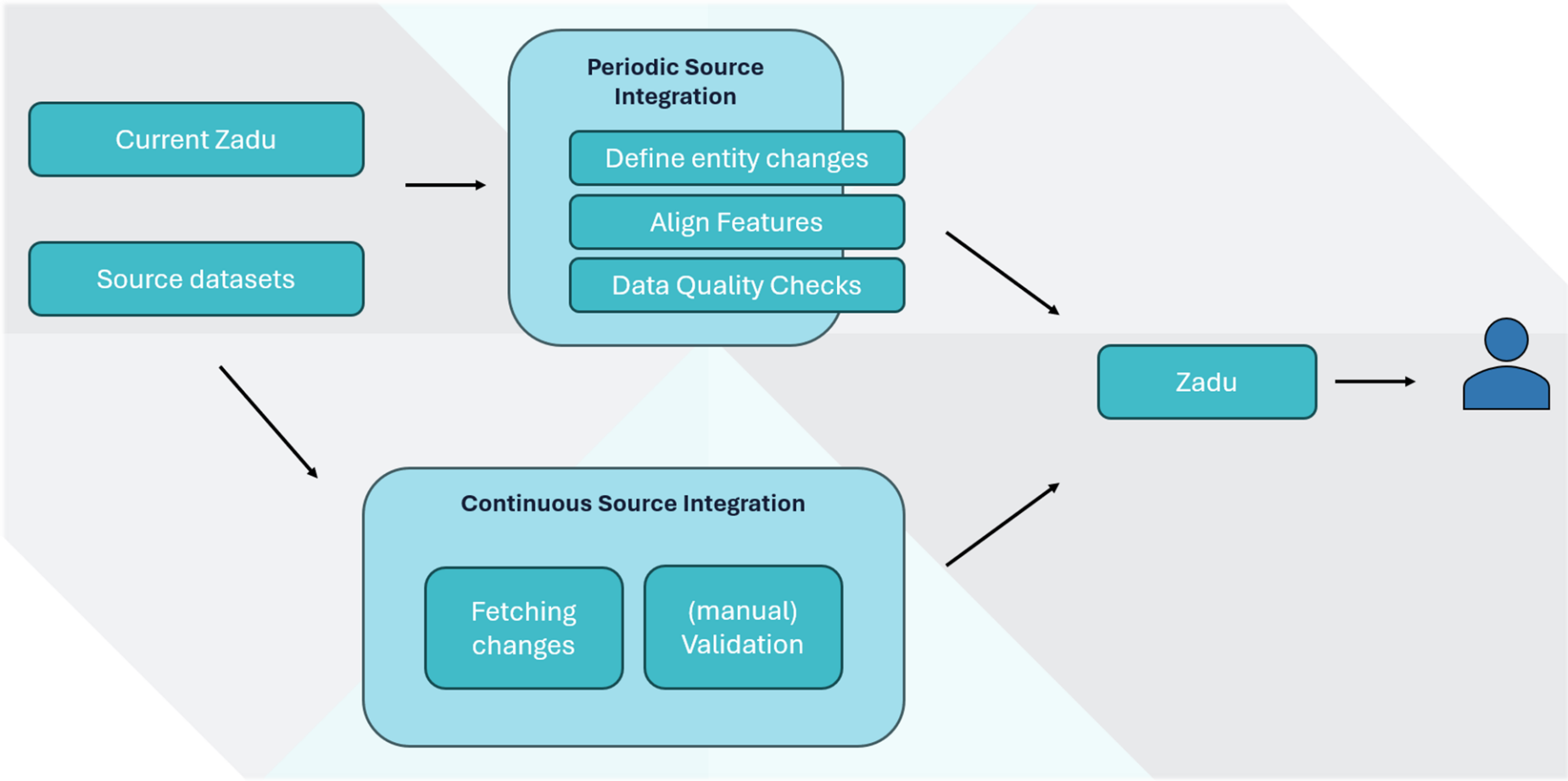
# Bound the workload.

Make country-scale processing infrastructure-independent.

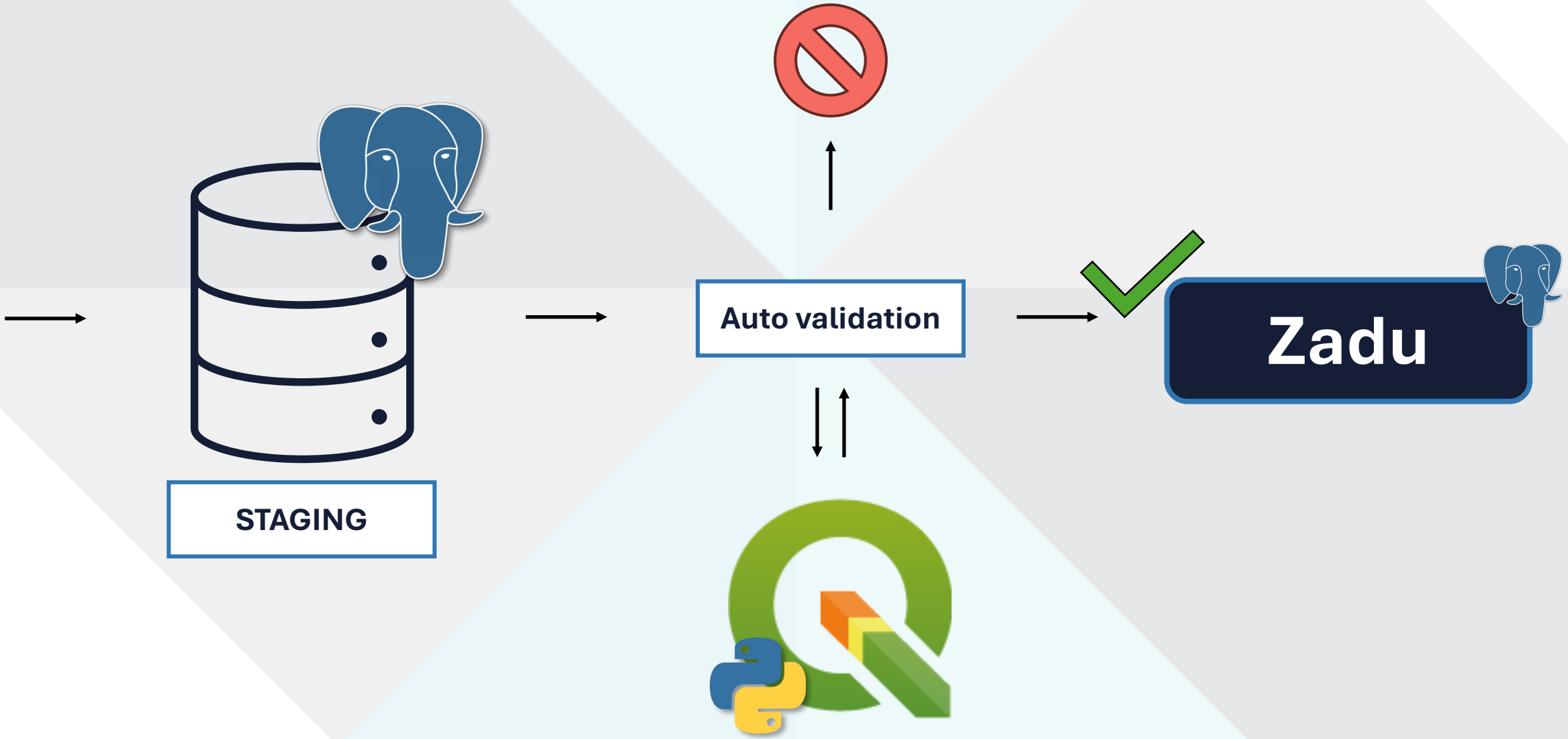




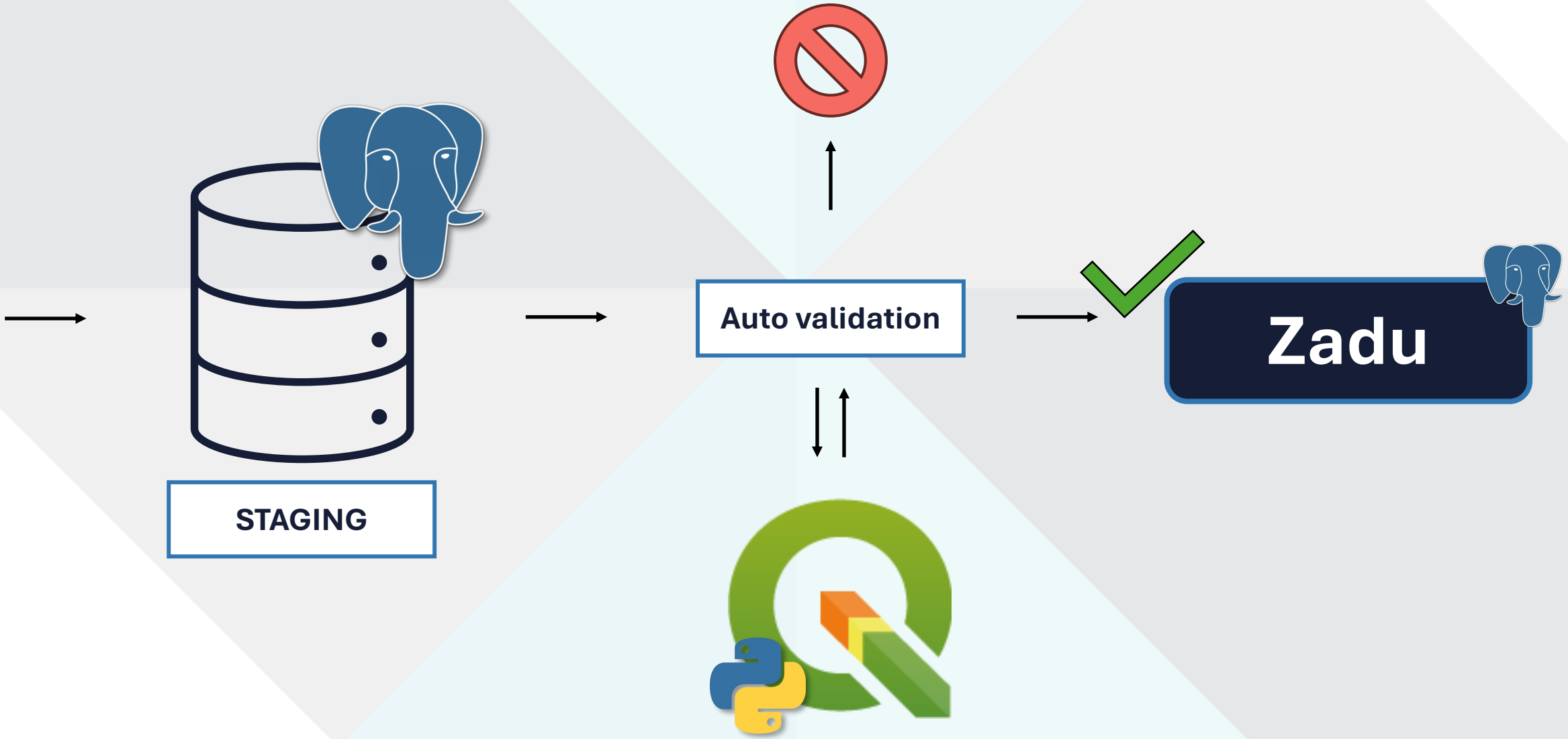

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Zadu


Current Zadu
Source datasets
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Zadu

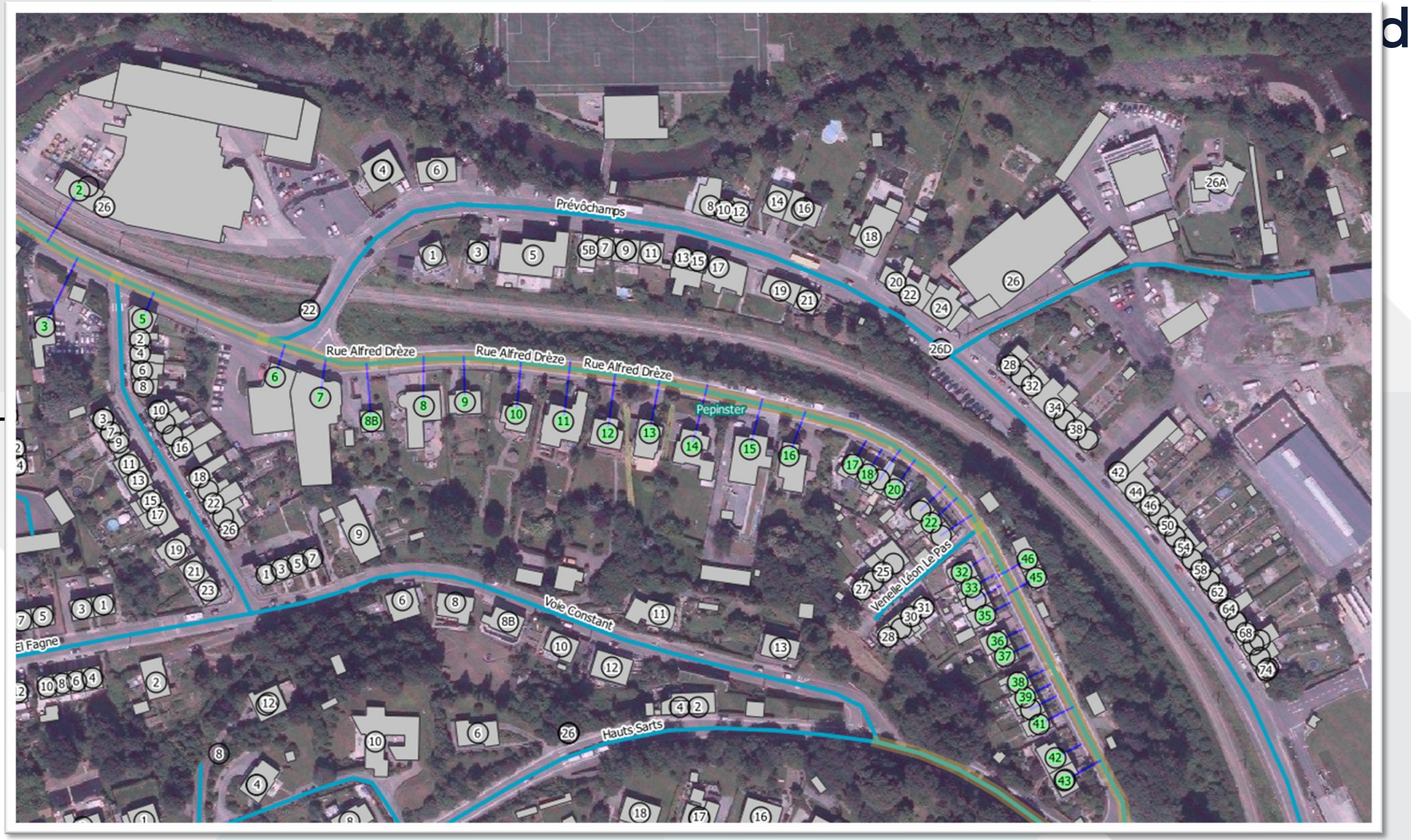
zadu

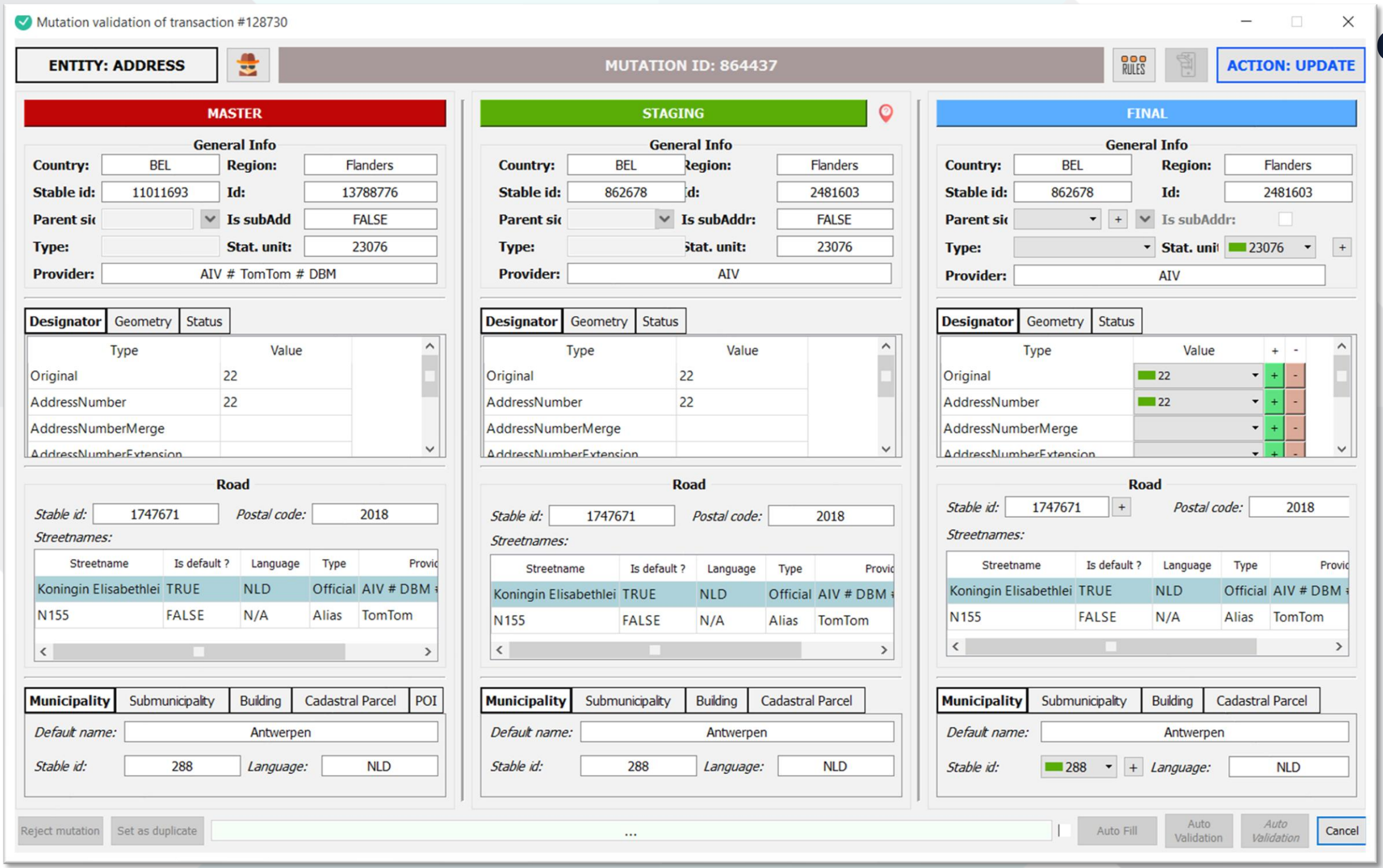


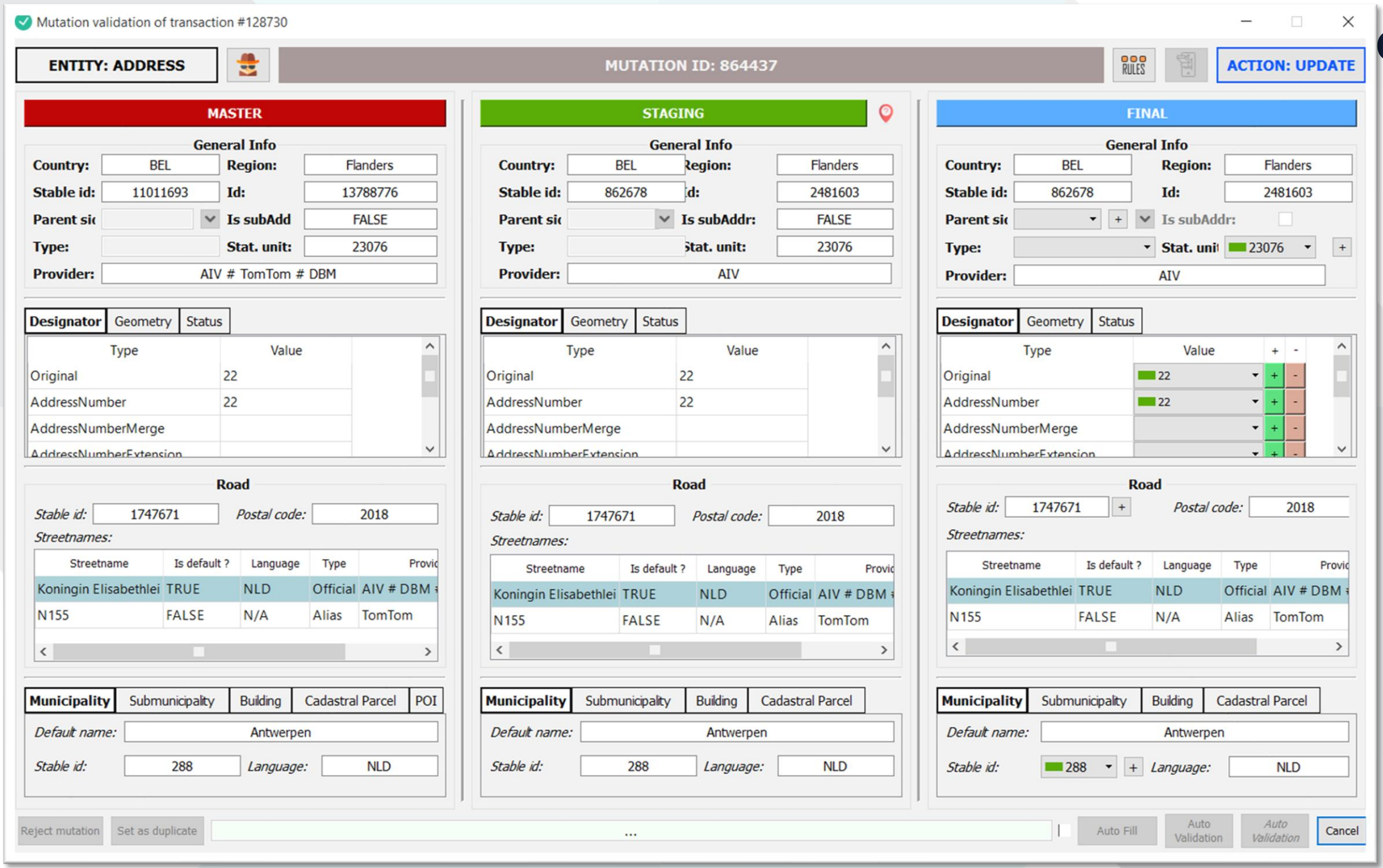
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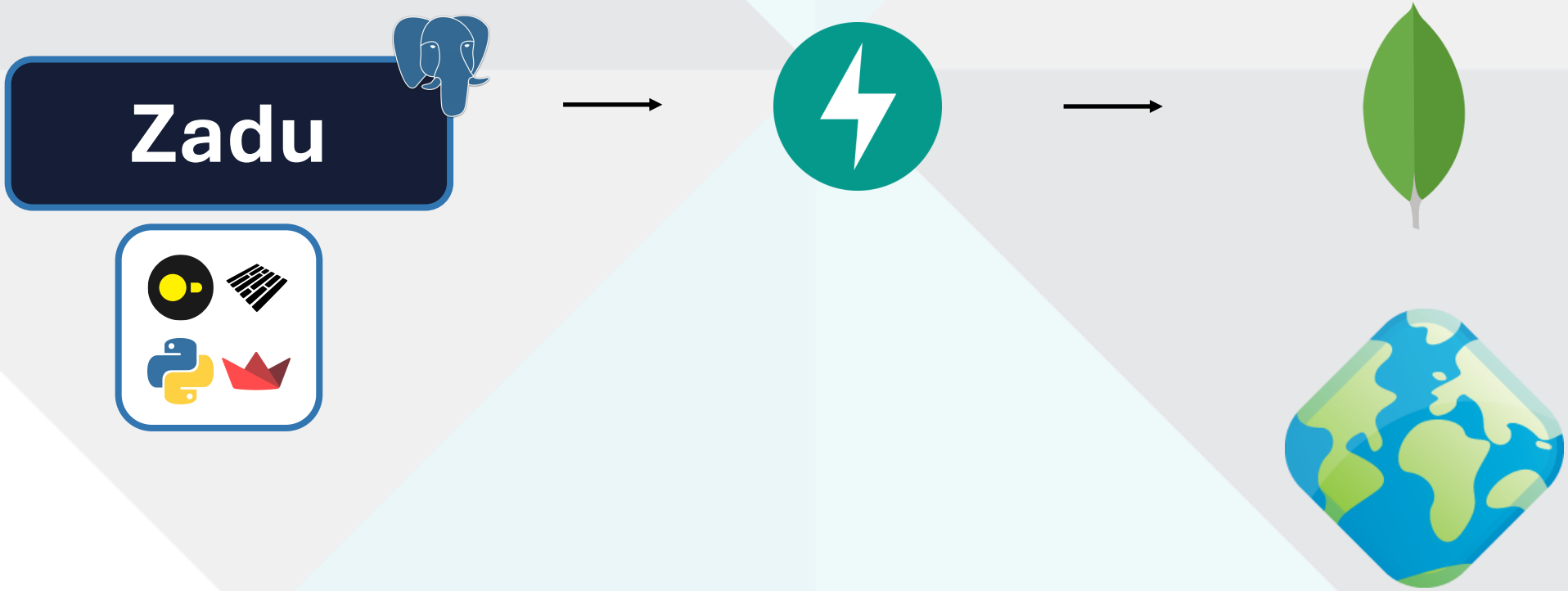


du


Prévôchamps
Rue Alfred Drèze
Rue Alfred Drèze
Rue Alfred Drèze
Pepinster
26D
Voie Constant
Hauts Sarts
Vieille Léon Le Pas
El Fagne


Mutation validation of transaction #128730
ENTITY: ADDRESS
MUTATION ID: 864437
RULES
ACTION: UPDATE
MASTER
STAGING
FINAL
General Info
Country: BEL
Region: Flanders
Stable id: 11011693
Id: 13788776
Parent sic
Is subAdd
FALSE
Type:
Stat. unit: 23076
Provider: AIV # TomTom # DBM
Stable id: 862678
Id: 2481603
Is subAddr:
Stat. unit: 23076
Provider: AIV
Designator
Geometry
Status
Type
Value
Original 22
AddressNumber 22
AddressNumberMerge
AddressNumberExtension
Road
Stable id: 1747671
Postal code: 2018
Streetnames:
Streetname
Is default ?
Language
Type
Provic
Koningin Elisabethlei TRUE NLD Official AIV # DBM
N155 FALSE N/A Alias TomTom
Municipality
Submunicipality
Building
Cadastral Parcel
POI
Default name: Antwerpen
Stable id: 288
Language: NLD
Reject mutation
Set as duplicate
Auto Fill
Auto Validation
Cancel


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**Build with open source. Scale with smart architecture.**



**Jeroen Stiers**
Product Owner, Zadu

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A Merkator Company